

EART 26001
Energy and Resources

Dr Stefan Schroeder

Lecture 6-1: Gold

Energy and Resources
Learning goals week 6

This week you will learn to:

- **Describe the uses and physical and chemical properties of gold, and how they contribute to formation of gold deposits.**
- **Recognize the geological processes forming placer and hydrothermal (orogenic) gold deposits.**
- **Describe the methods for gold exploration and production, and recognize the environmental and societal impact of gold mining.**



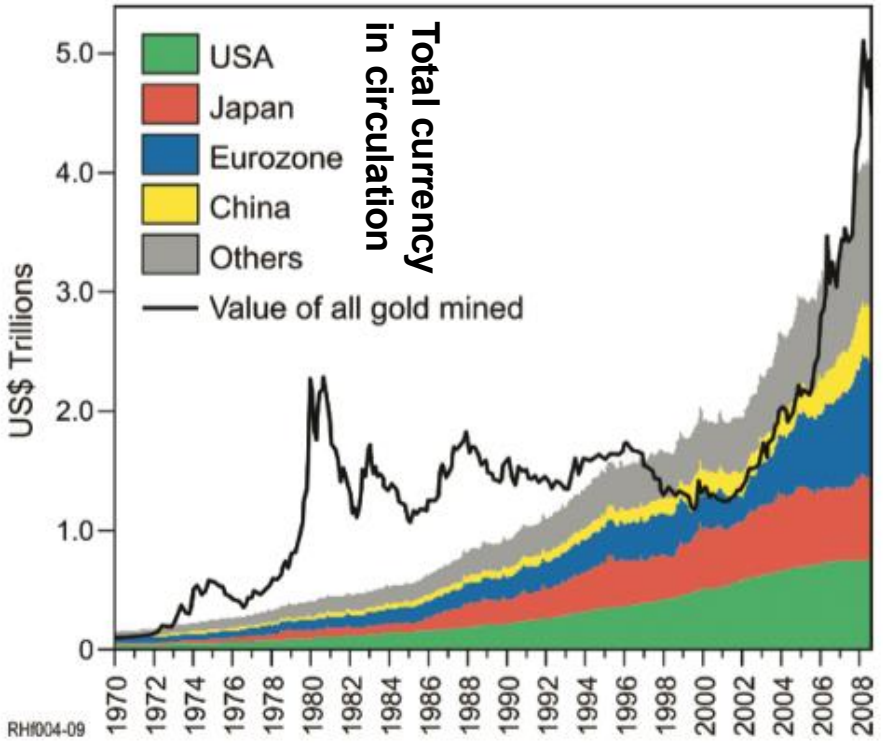

My precious



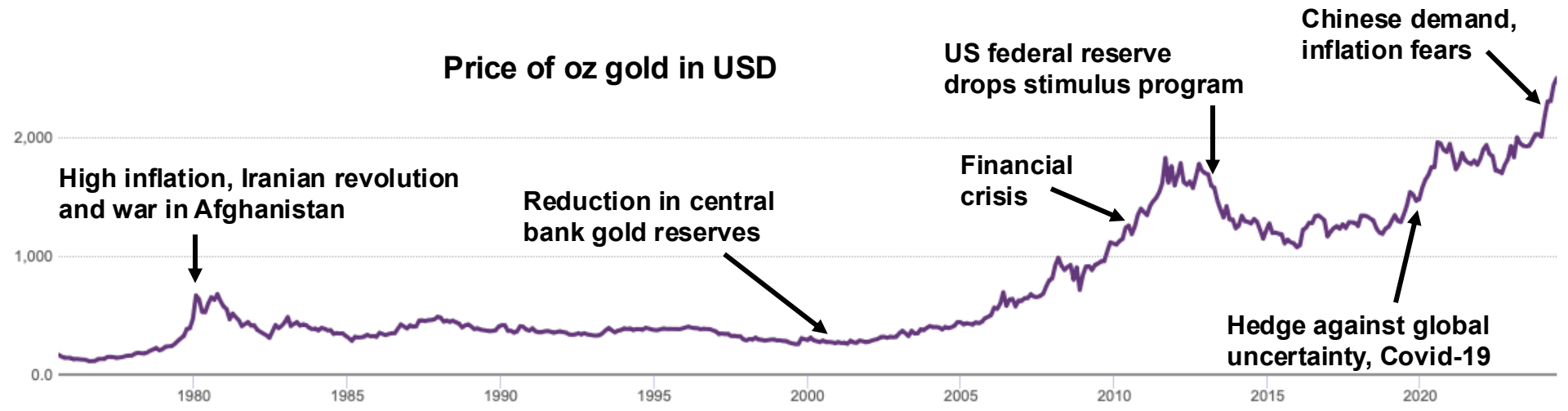

KRUGERRAND
19
86
FYNGOUD 1OZ FINE GOLD
umicore
FEINGOLD
999.9
1000g

Introduction
Value of gold

”Gold is valuable, because we, the human race, continue to believe it is so, whatever the reason” (Butt 2009)



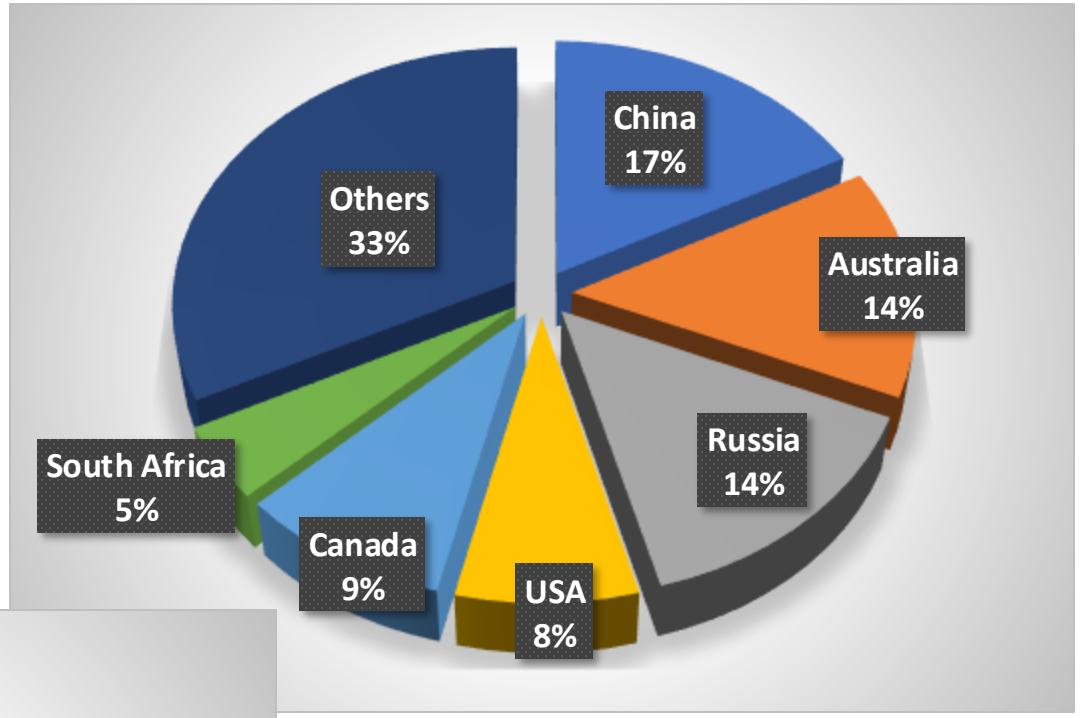
Butt 2009



© World Gold Council

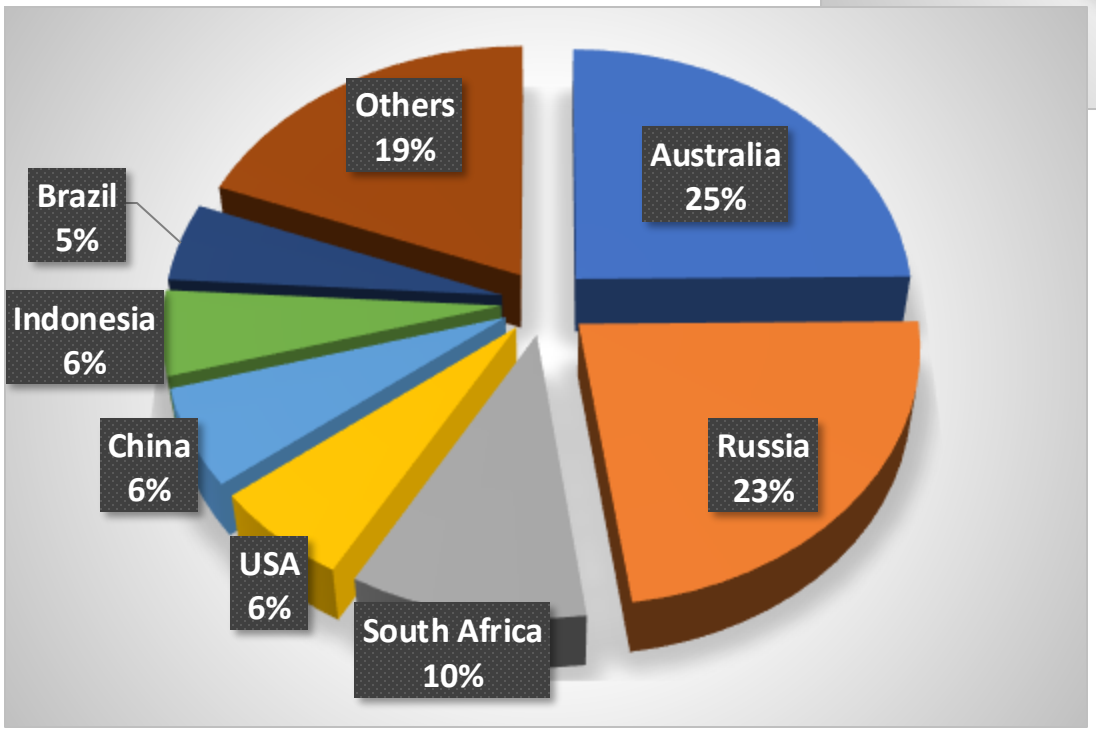
Introduction
Reserves

Au Mine production 2023



Source : USGS

Au Reserves 2023



Source : USGS

Introduction
Properties of gold

Symbol	Au (Latin *aurum*)
Colour	Gold
Atomic number	79
Atomic mass	196.96655 amu
No. neutrons	118
Electrons per shell	2, 8, 18, 32, 18, 1
Abundance	Earth's crust: ~1.3 ppb Iron meteorites: 1 ppm Seawater: 0.005–0.01 ppb
Isotopes	***amu*** ***half-life*** 194 1.6 days 195 186.1 days 195m 30.5 seconds 196 6.2 days 197 stable 198 2.7 days 199 3.14 days
Crystal system	Face-centred cubic, close packed
Unit cell	0.40783 nm
Melting point	1064.43°C
Boiling point	2807.0°C
Density at 293 K **Liquid density**	19.32 g/cm^3 17.31 g/cm^3
Hardness	Mohs: 2 Vickers HV: 25
Thermal conductivity	317 W m^{-1} K^{-1}
Electrical resistivity at 20°C	0.022 micro-ohm m
Young's modulus	79
Poisson's ratio	0.42
0.2% proof stress	30 MPa
Tensile stress	124 MPa
Malleability	Foil thickness: 0.1–0.125 μm
Ductility	Wire thickness: <25 μm

Profitable @1-10ppm, i.e. factor 770-7700

siderophile

Butt 2009

Introduction
Properties of gold

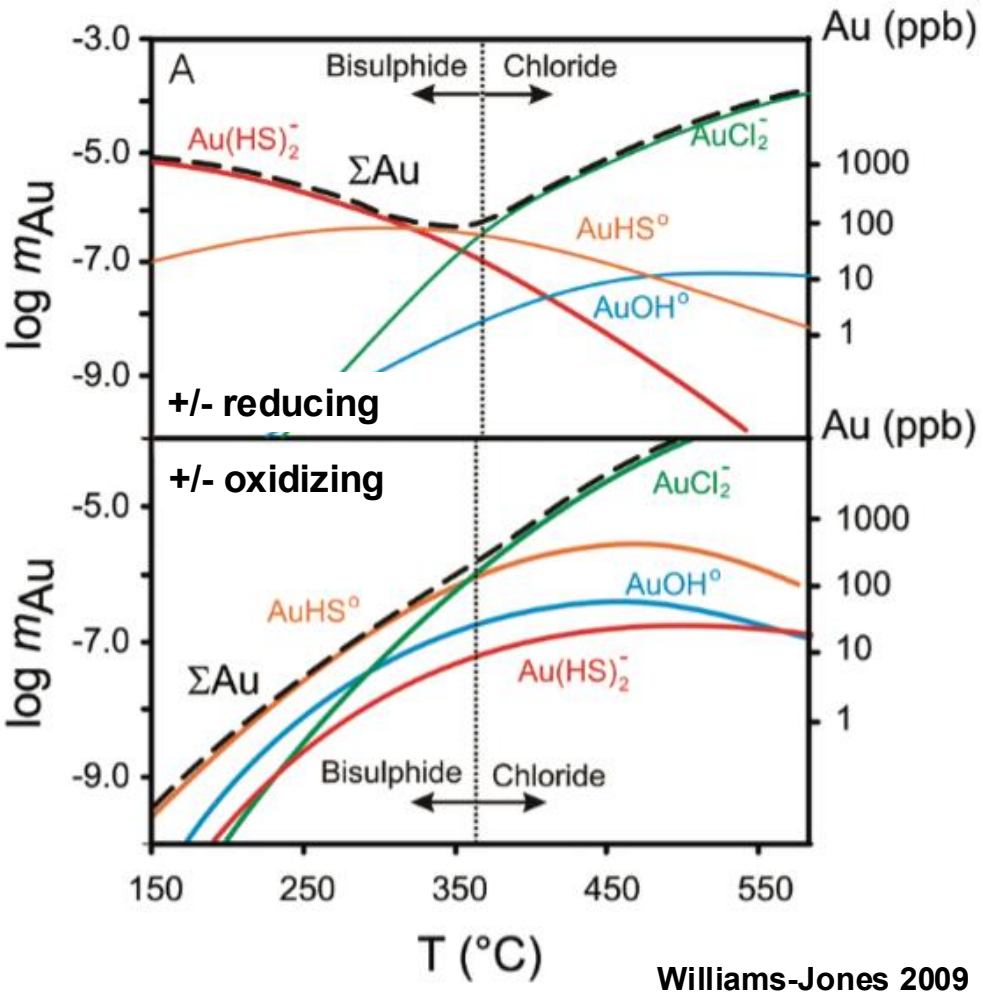
Reaction at ambient temperatures:
with aqua regia, CN^-, $S_2O_3^{2-}$

Reaction at high temperatures:

$$Au_S + HS^- + H^+ + \tfrac{1}{4}O_2 \leftrightarrow Au(HS)^0 + \tfrac{1}{2}H_2O$$

$$Au_S + 2Cl^- + H^+ + \tfrac{1}{4}O_2 \leftrightarrow AuCl_2 + \tfrac{1}{2}H_2O$$

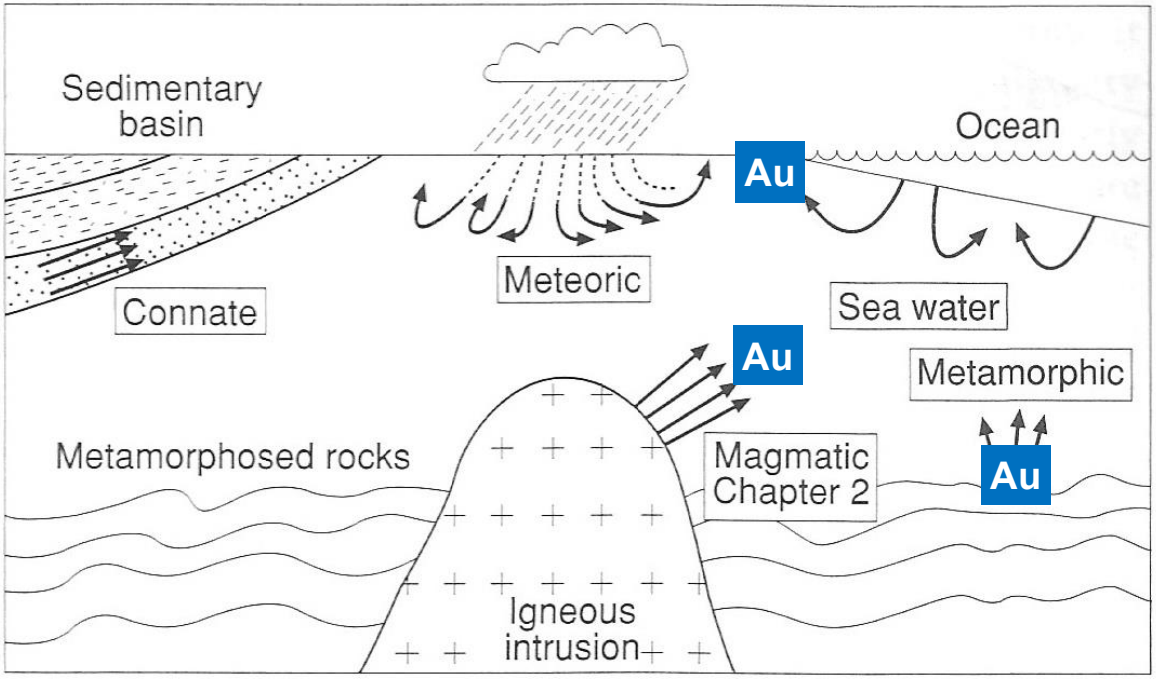
Gold solubility vs temperature



Williams-Jones 2009

Gold
Deposit types

- **Magmatic-hydrothermal fluids:**
  - **Epithermal deposits**
  - **By-product in volcanogenic massive sulfides and porphyries**
- **Hydrothermal fluids:**
  - **Orogenic deposits (can involve magmatic-hydrothermal fluids)**
- **Meteoric sedimentary fluids:**
  - **Alluvial placers**



Robb 2005

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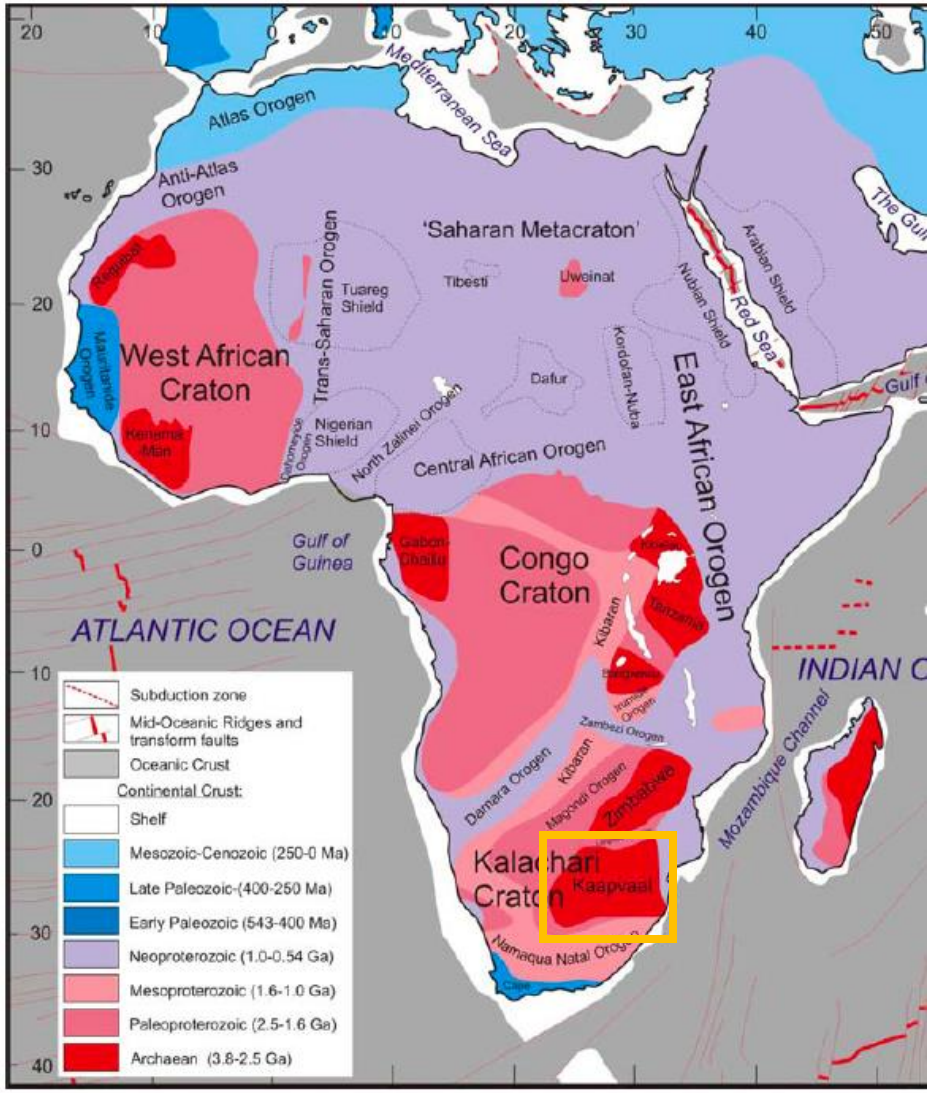
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Lecture 6-2: Hydrothermal (orogenic) gold deposits (Barberton)



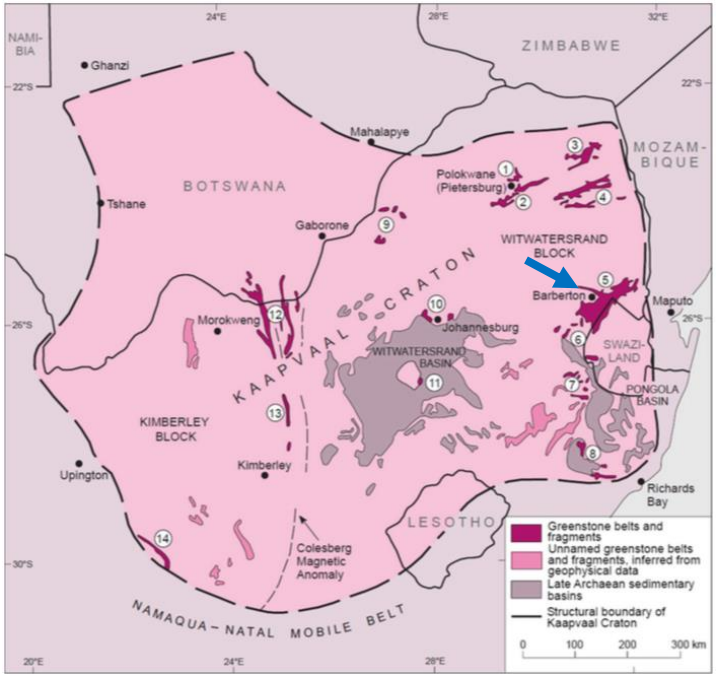
Hough 2009

Barberton greenstone belt
Setting

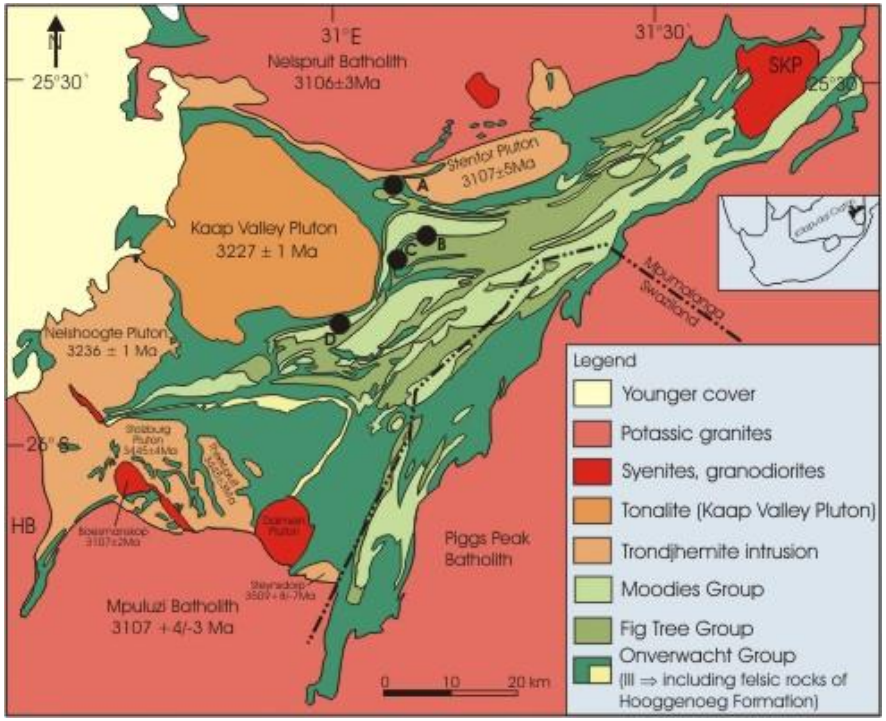


Van Hinsbergen 2011, after Gubanov & Mooney 2009

Brandl 2006

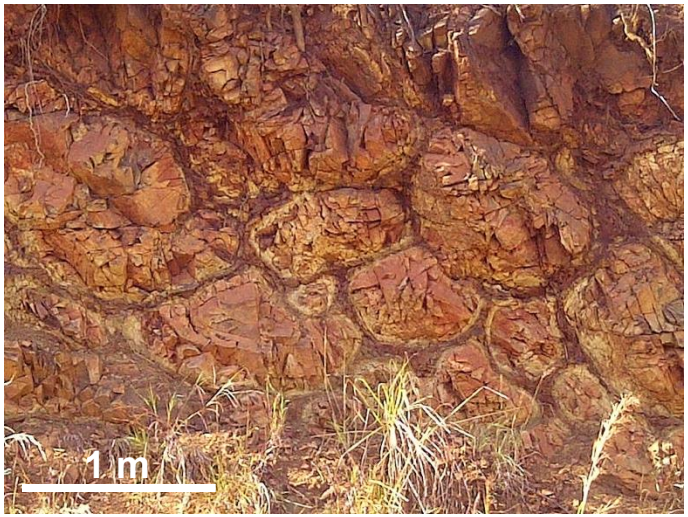


Rowan 2007

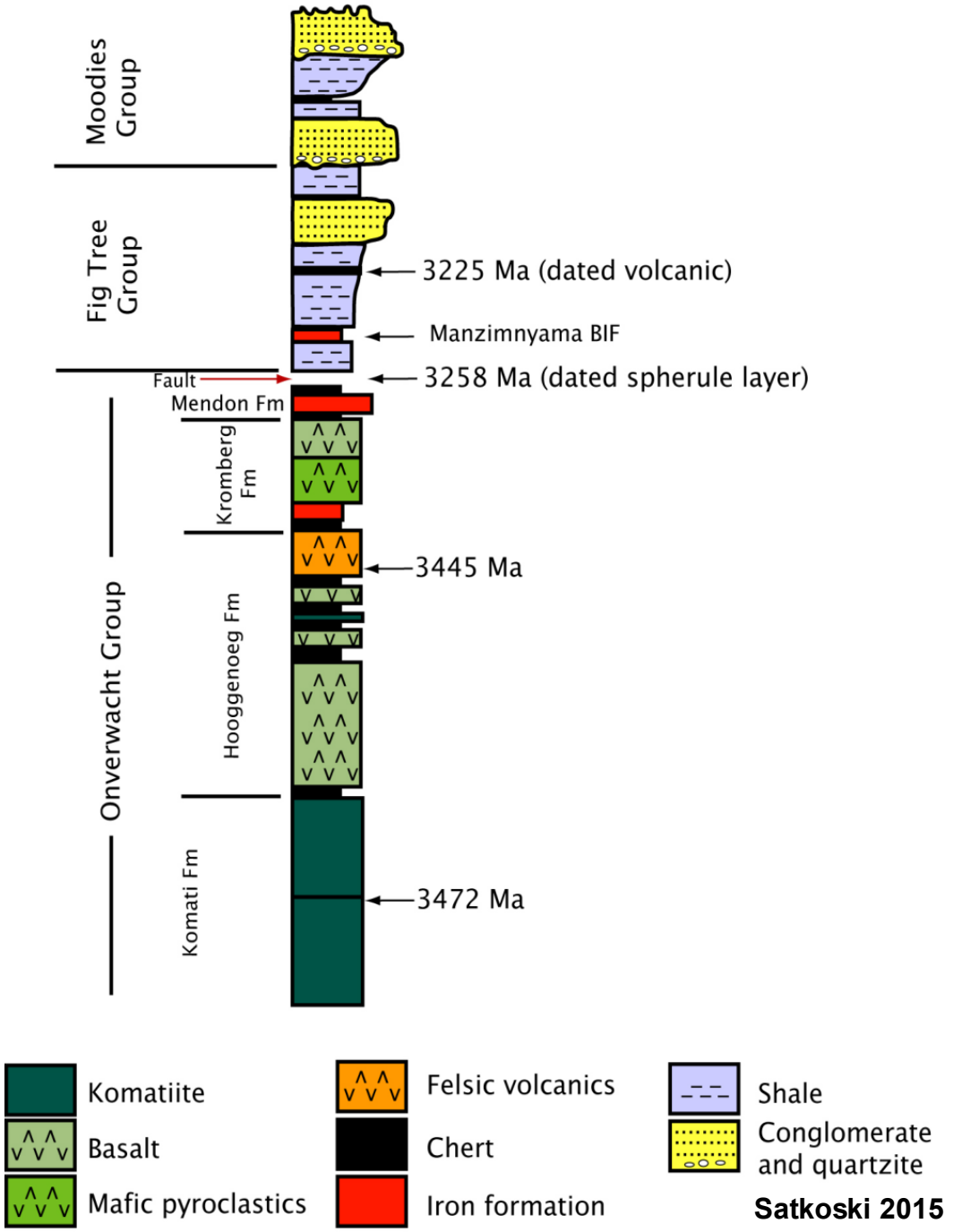


Barberton greenstone belt
Stratigraphy



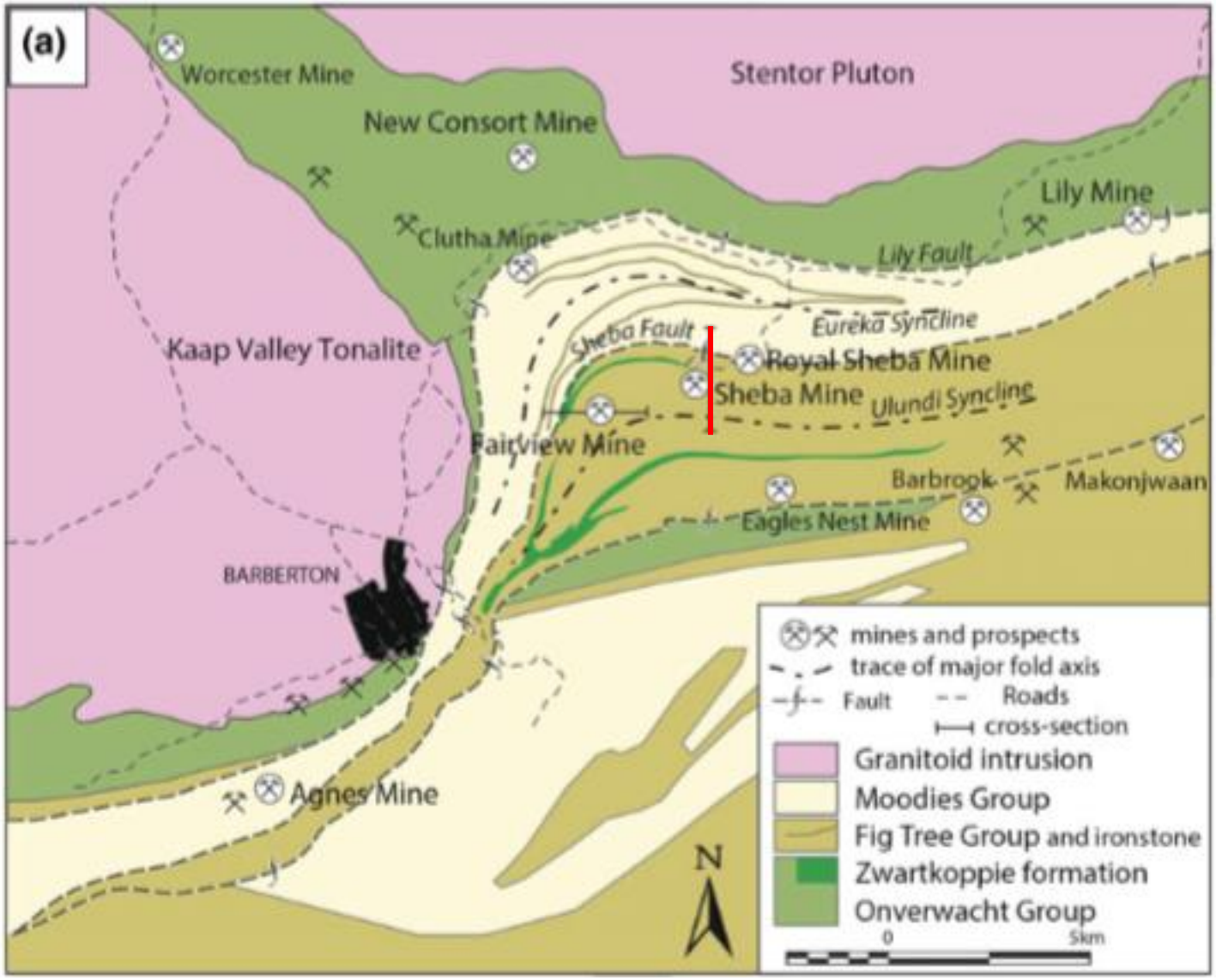


© Barberton WHS



Barberton gold deposits
Setting

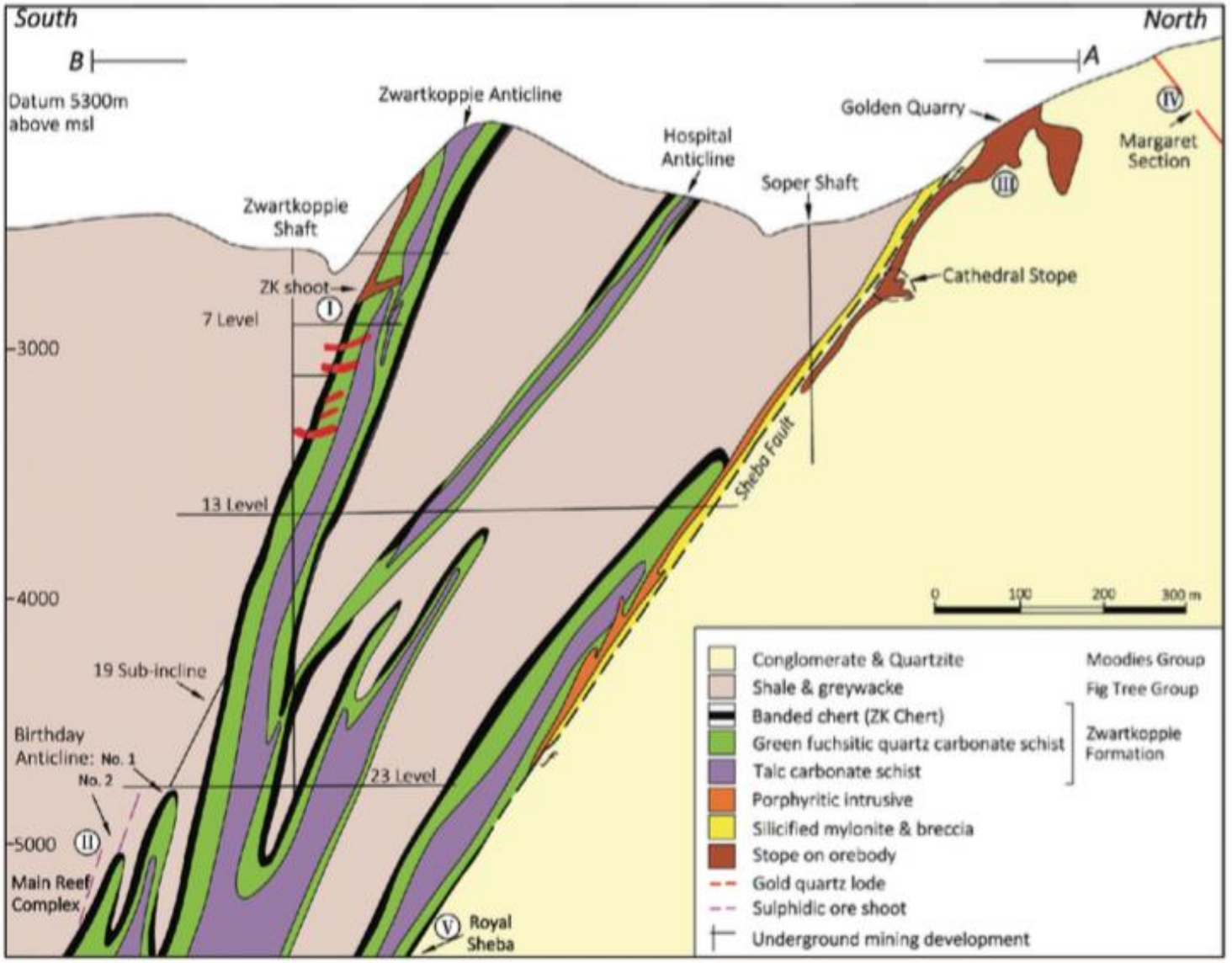
Major gold mines around Barberton



Agangi 2019

Barberton gold deposits
Structure

Structural context of deposits



Pearton 2017

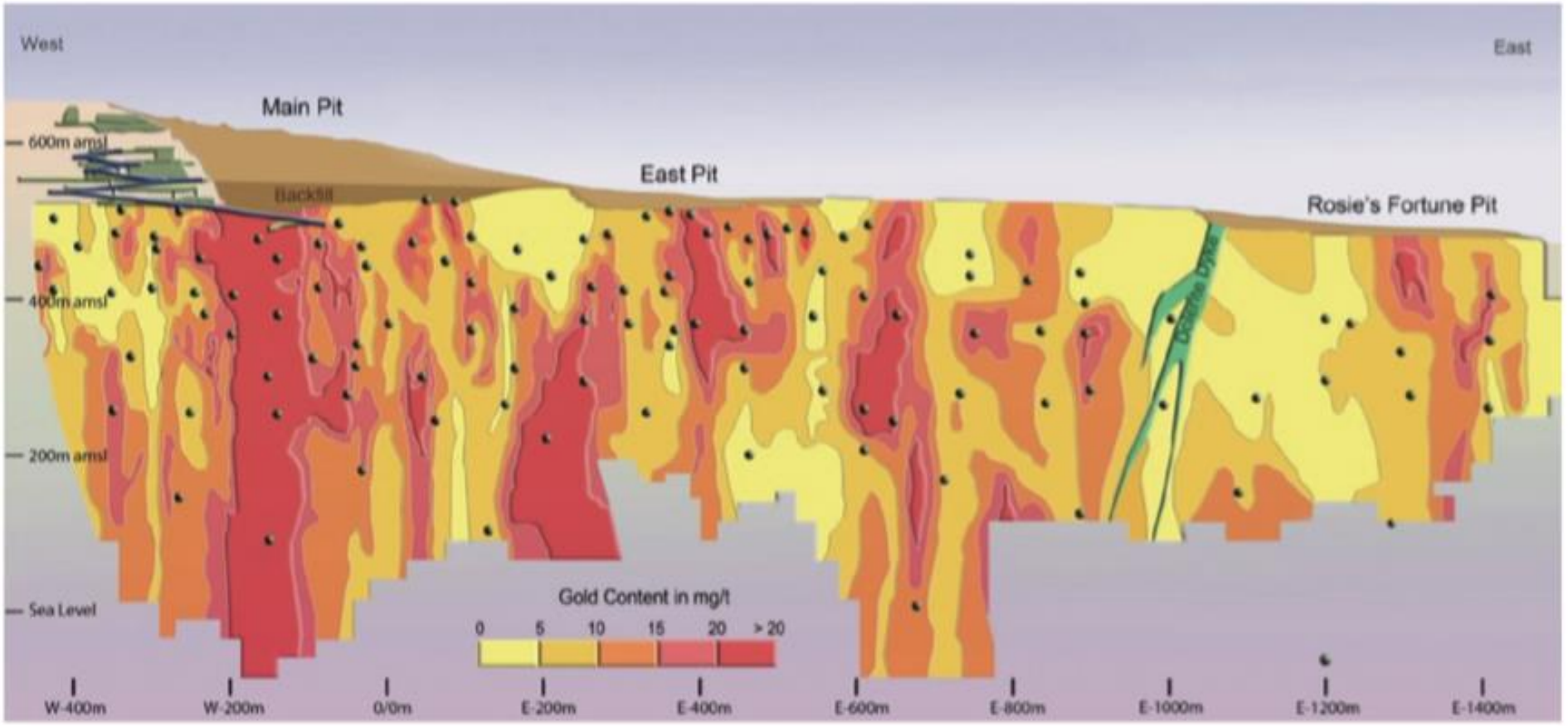
Barberton gold deposits
Structure



Pearton 2017

Barberton gold deposits
Gold content

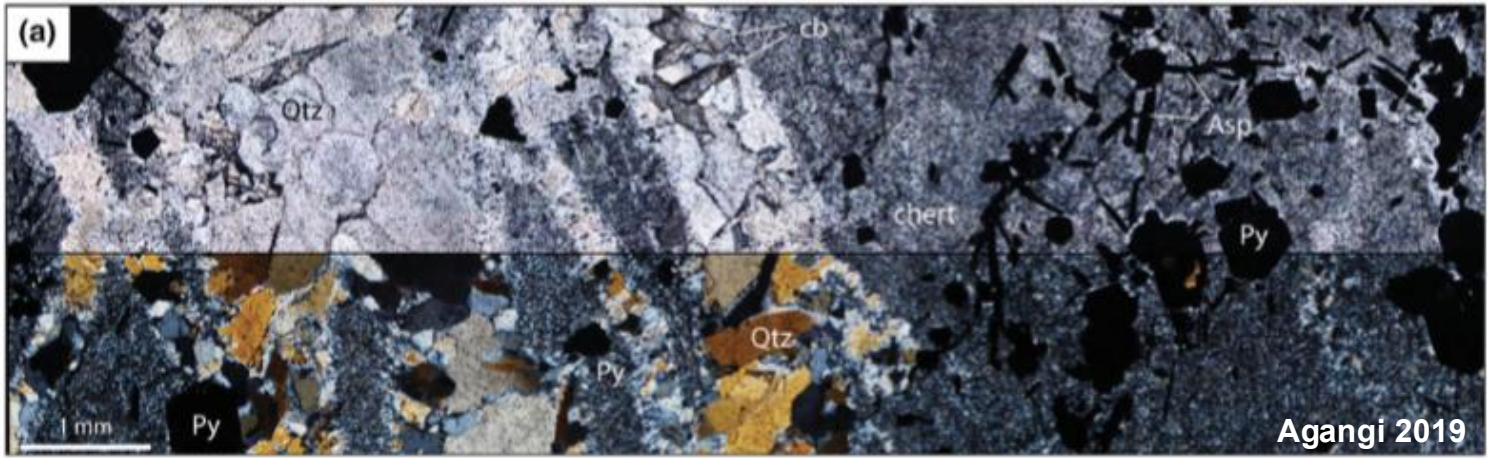
Lily Mine



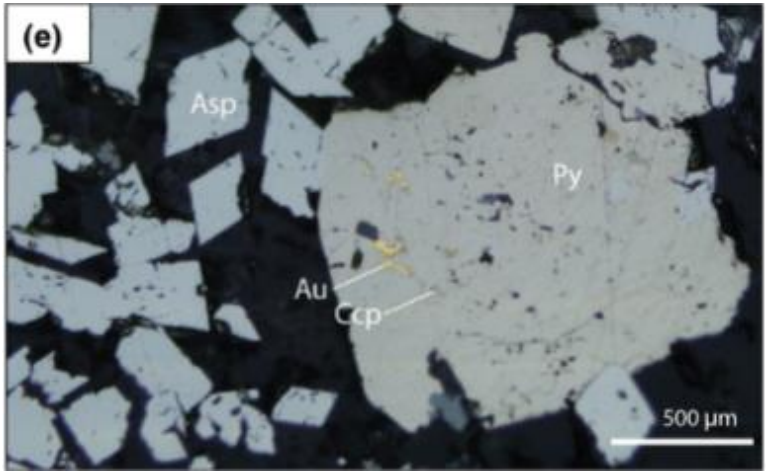
- **Veins 100's m long, <1 m wide**
- **Disseminations in country rock**

Pearton 2017

Barberton gold deposits
Mineralogy





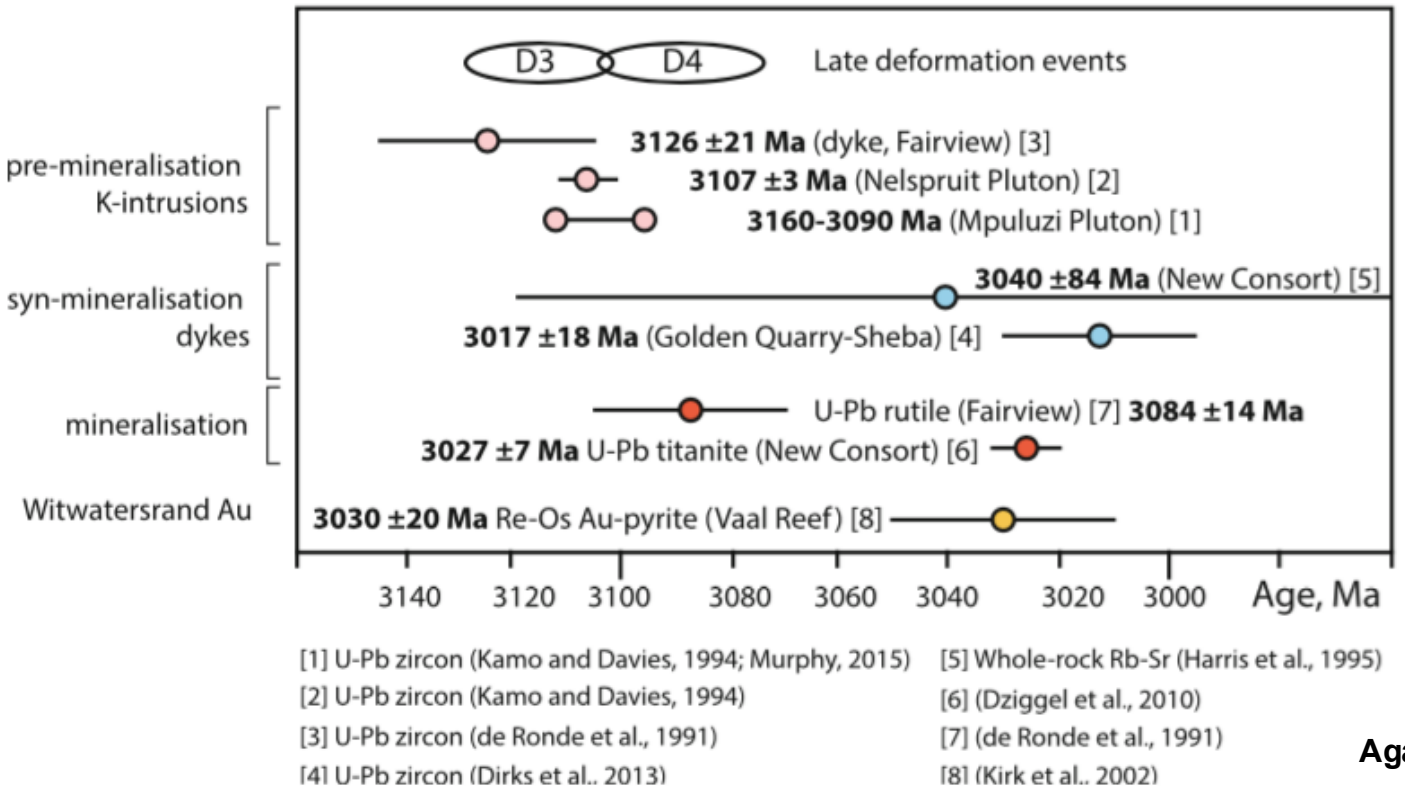


Agangi 2019

- Quartz-carbonate veins with pyrite, chalcopyrite, sphalerite, As-Ni sulfides
- Gold occurs as free gold, inclusions in sulfide minerals, and as very fine, invisible (refractory) gold
- Grade <60g/t, average 8g/t

Barberton gold deposits
Genesis

Age distribution

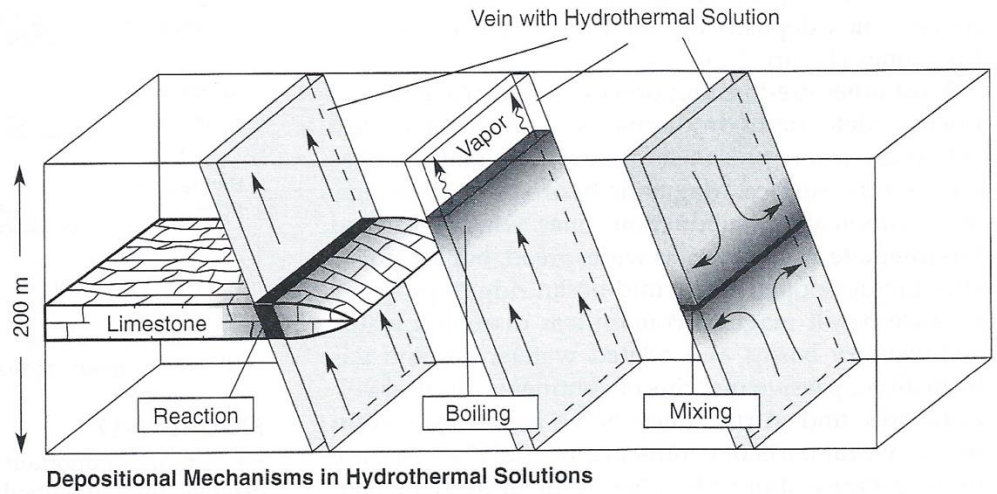


[1] U-Pb zircon (Kamo and Davies, 1994; Murphy, 2015)
[2] U-Pb zircon (Kamo and Davies, 1994)
[3] U-Pb zircon (de Ronde et al., 1991)
[4] U-Pb zircon (Dirks et al., 2013)
[5] Whole-rock Rb-Sr (Harris et al., 1995)
[6] (Dziggel et al., 2010)
[7] (de Ronde et al., 1991)
[8] (Kirk et al., 2002)

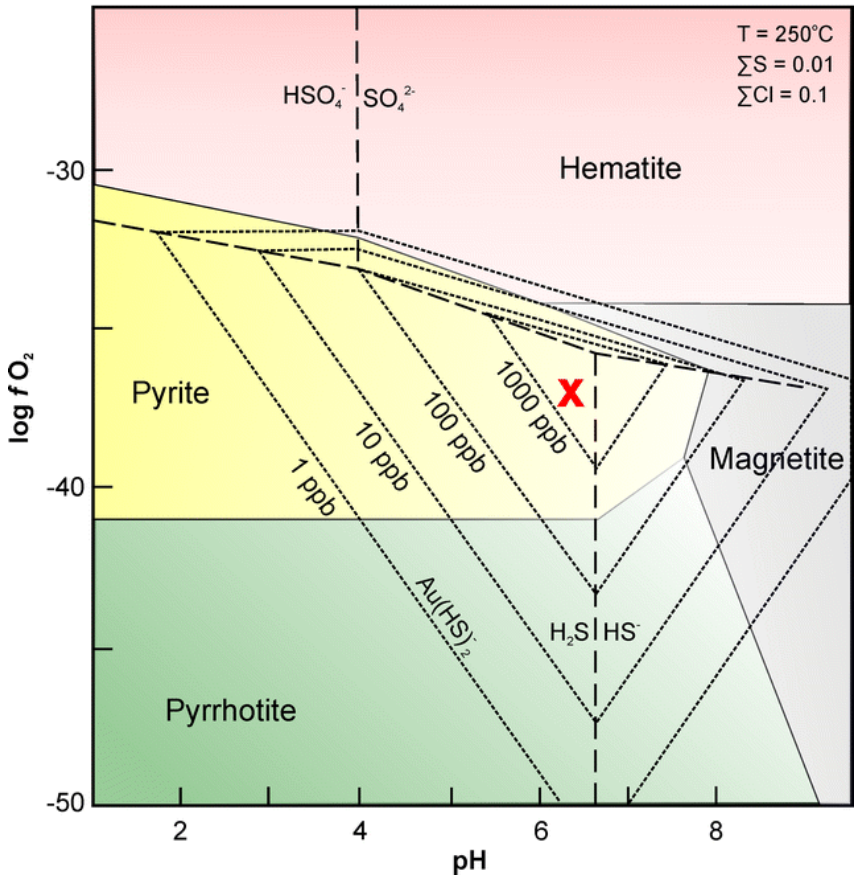
Agangi 2019

- **Fluids: low salinity, H_2O-CO_2 fluids, from 10-20 km depth**
- **Devolatilization of water-bearing minerals (chlorite, carbonates) during regional metamorphism: supracrustal rocks, (felsic intrusions, mantle?)**

Barberton gold deposits
Genesis



Kesler 1994

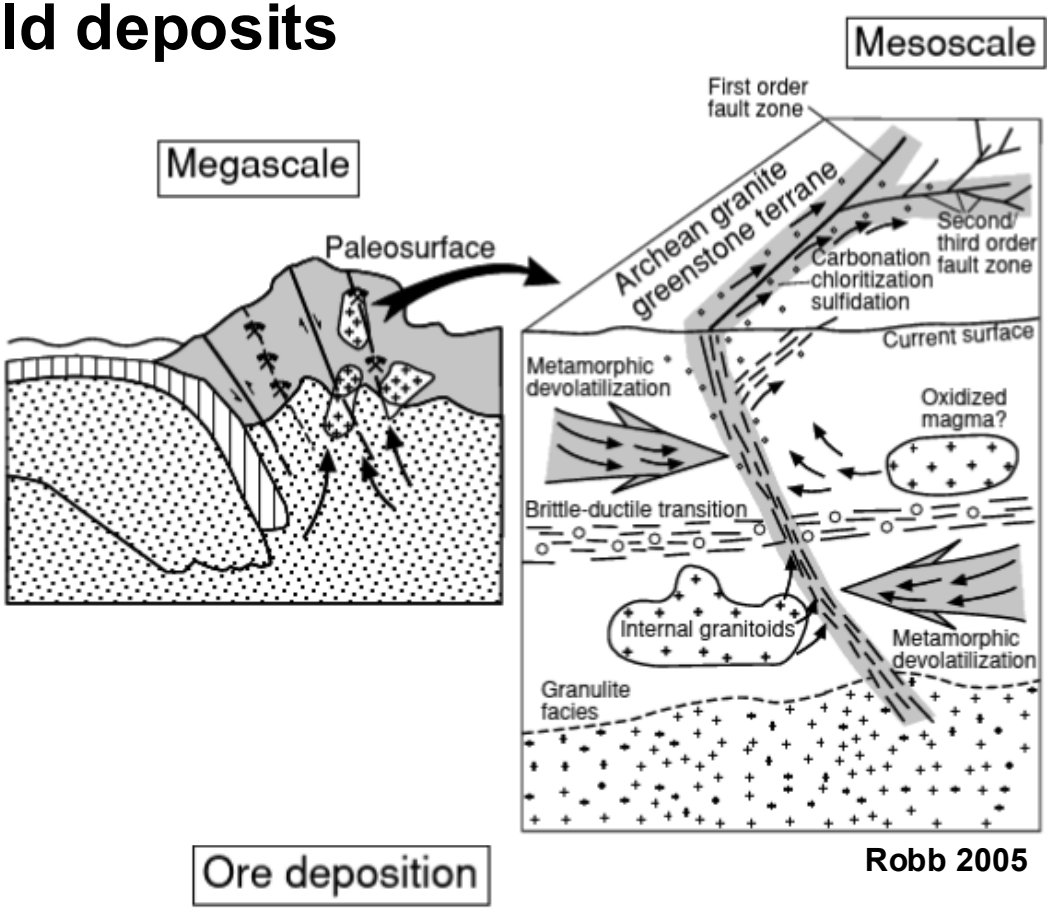


Frimmel 2014

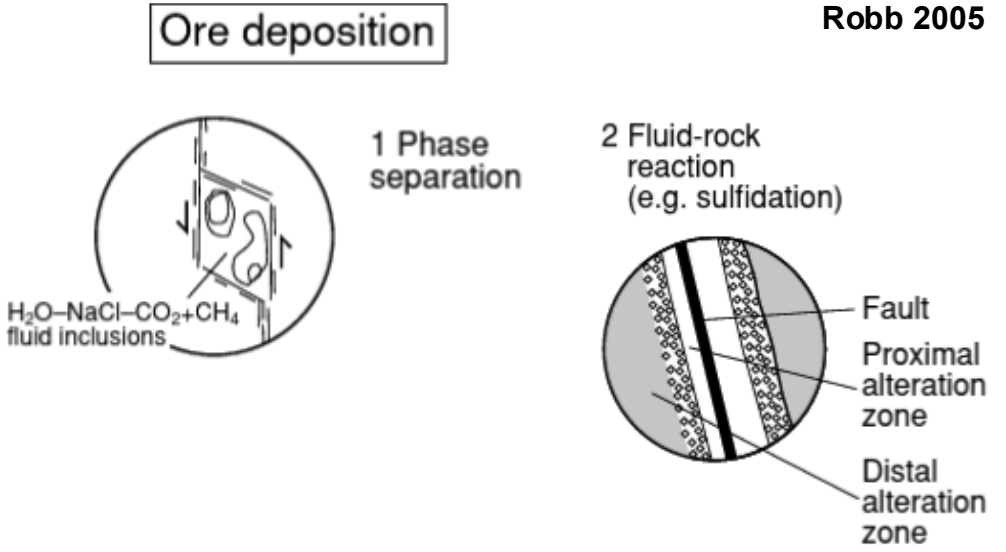
- Gold transported as AuHS and $AuHS_2$ in hydrothermal fluid, possibly from devolatilization of metamorphic rocks during orogenesis ("orogenic gold deposits")
- Precipitation via wall rock alteration, boiling, mixing with meteoric water, temperature drop

Hydrothermal (orogenic) gold deposits
Summary

- Hydrothermal fluids are major driver for metal transport and precipitation due to metals bonding with anions.
- Various sources – in orogenic gold deposits, hydrothermal fluids derive from breakdown of water-bearing minerals in metamorphosis and from igneous exsolution. Fluid transport along major crustal structures.
- Characterization of fluid source, composition and relation to regional and local structure are essential for exploration and deposit evaluation.
- Deposits known throughout Earth history, associated with all mountain belts (e.g. Andes – exploitation by Spanish empire made Spain a major power since 1500s).
- Local example: Cononish, Scotland – Paleozoic quartz veins in Proterozoic metamorphic rocks with 11g/t gold

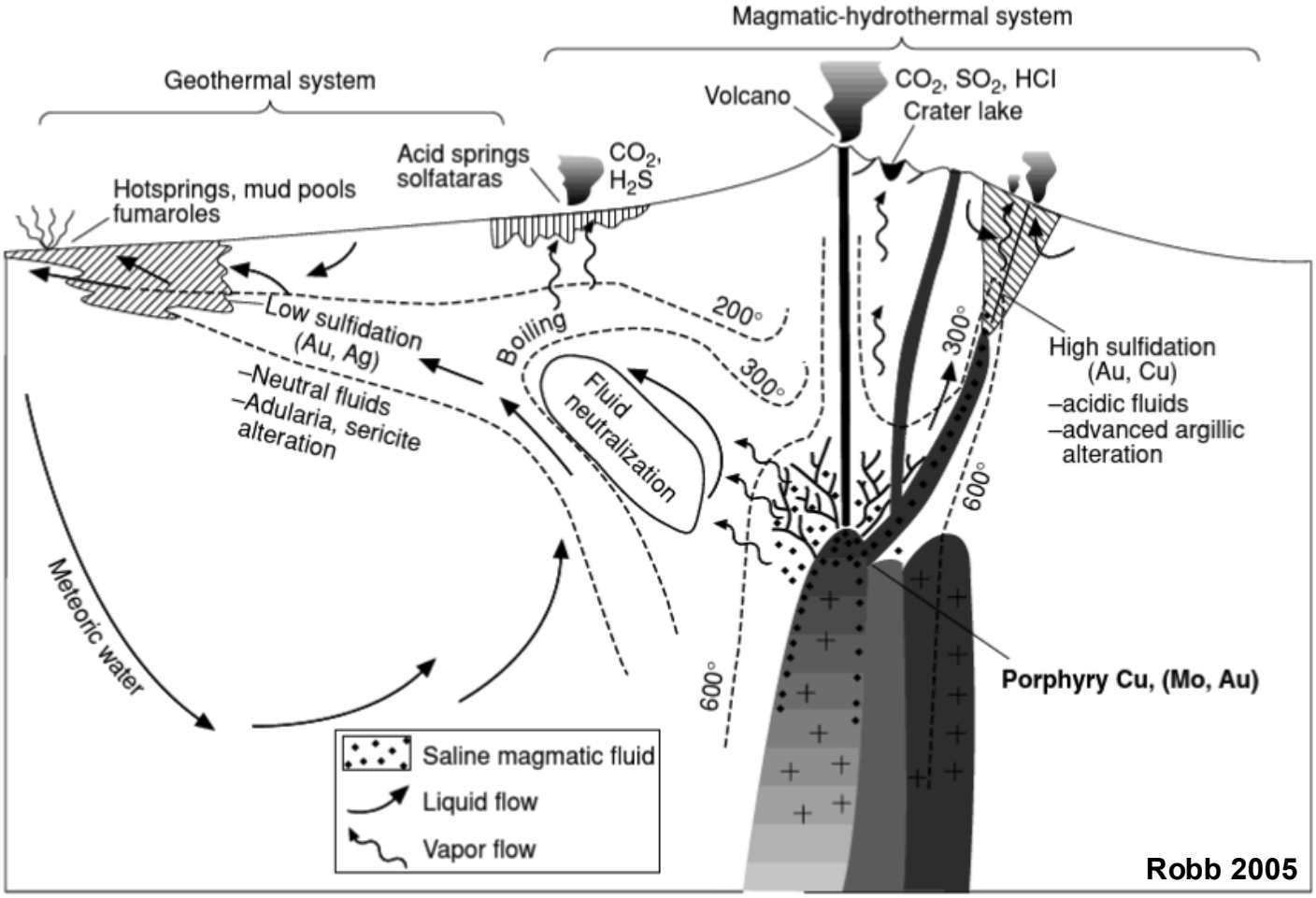


Robb 2005



Magmatic-hydrothermal gold deposits
Epithermal and porphyry deposits

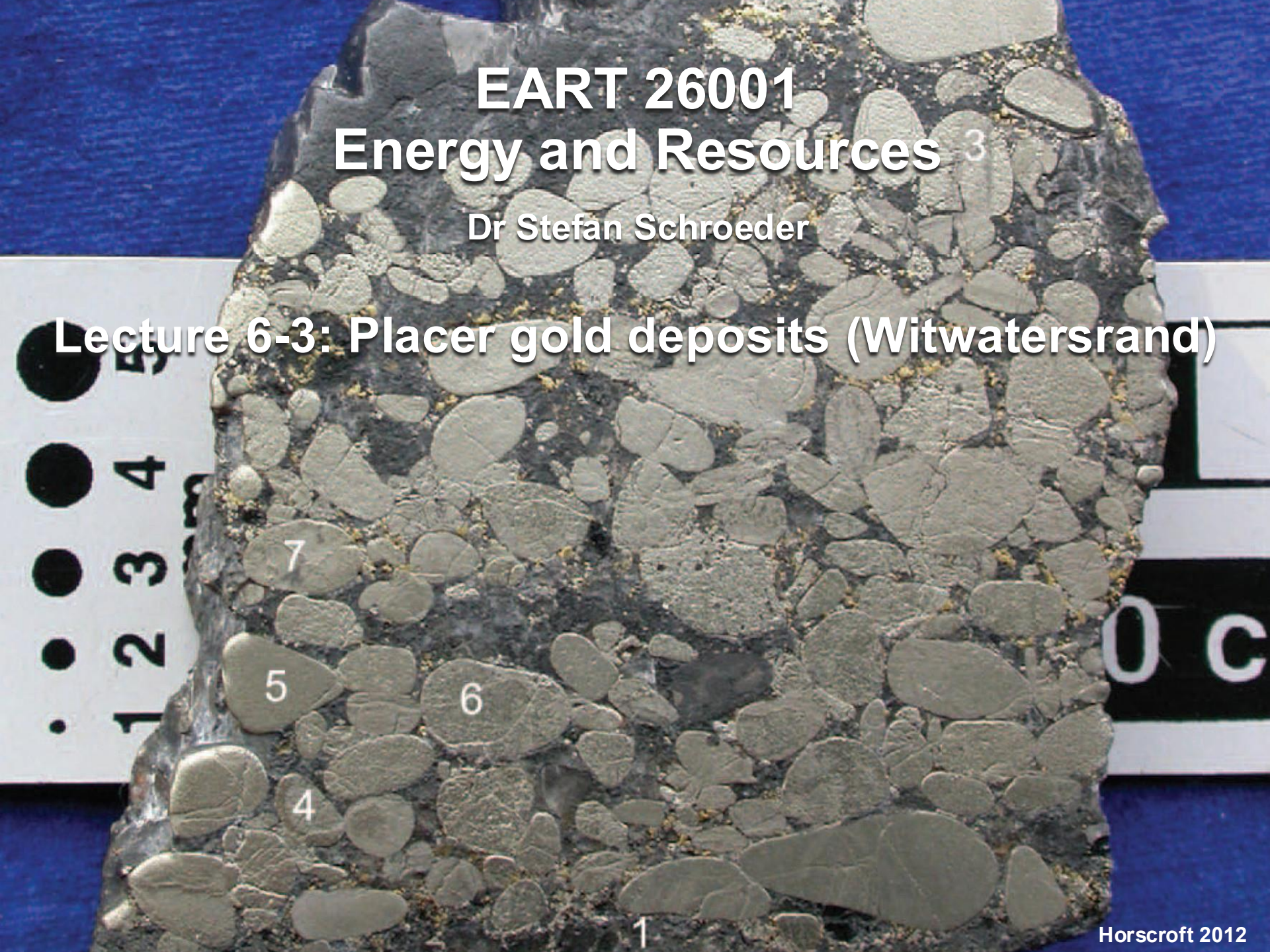
- Local fluid circulation around igneous intrusion involving magmatic-hydrothermal and heated meteoric (hydrothermal) water – epithermal (shallow crustal depth)
- Porphyries (mainly Cu +/- Mo, W, Au) form at depth from warmer magmatic-hydrothermal fluids (mesothermal)



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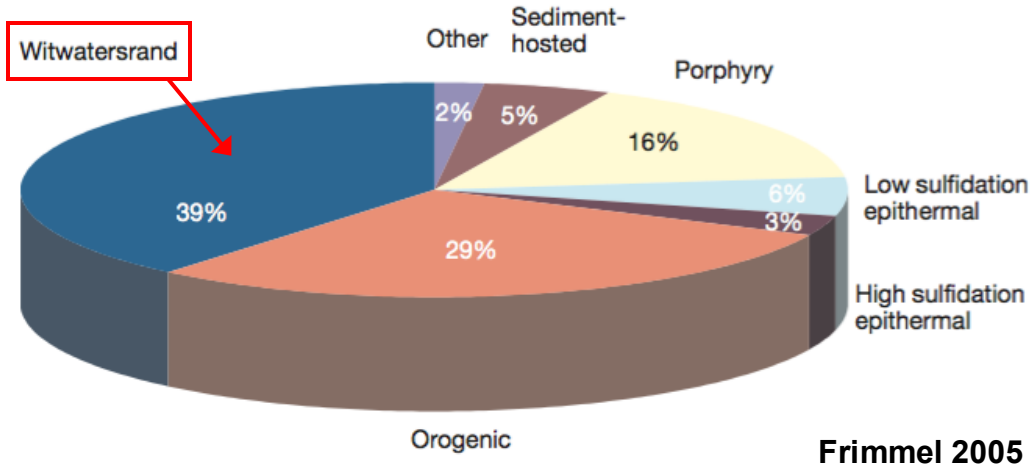
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Lecture 6-3: Placer gold deposits (Witwatersrand)



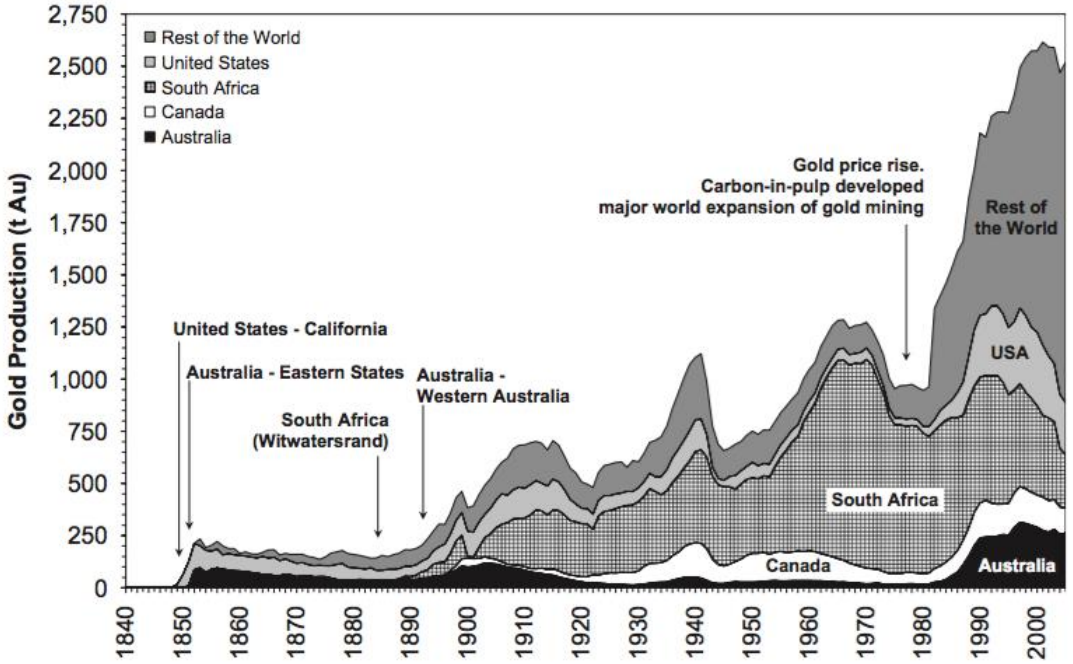
Horscroft 2012

Witwatersrand gold deposits
Context



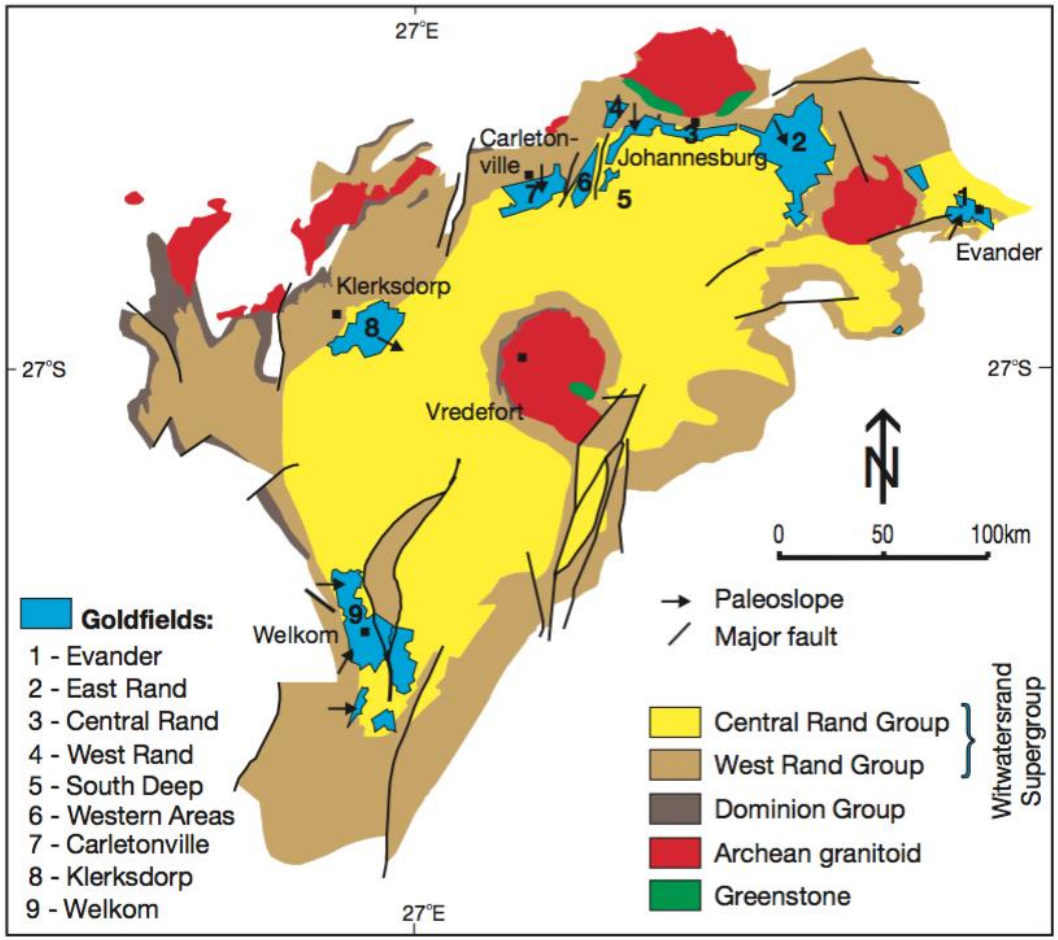
Frimmel 2005

- Witwatersrand gold fields have produced 50,000t = 40% of global gold mined 1886-2004
- Witwatersrand gold fields in 2004 had reserves of 38,000t = 46% of global reserves
- Today, the reserves have dropped to 6% of global reserves

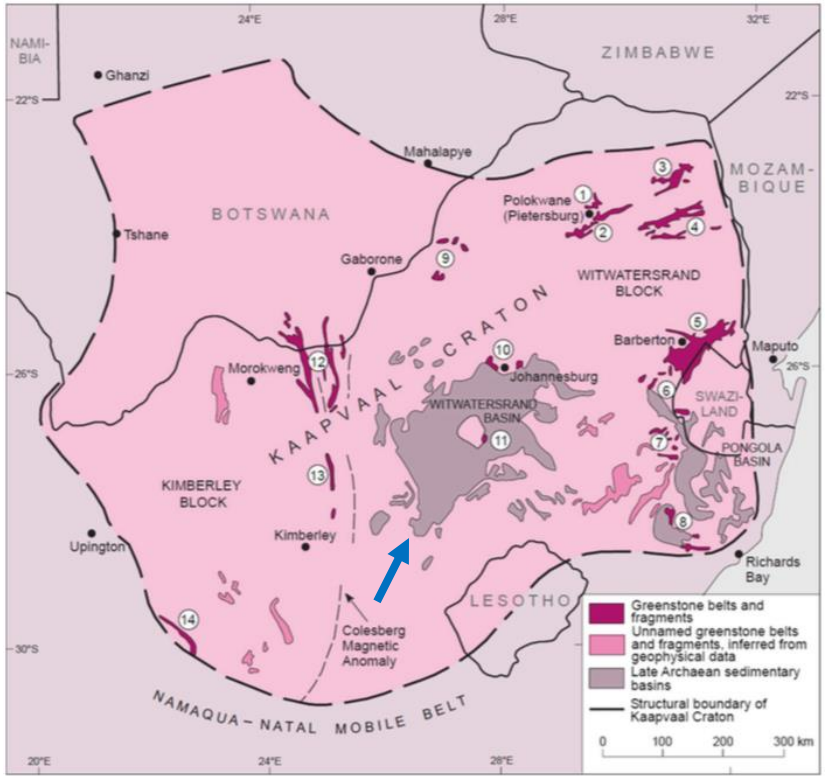


Mudd 2007

Witwatersrand gold deposits
Basin

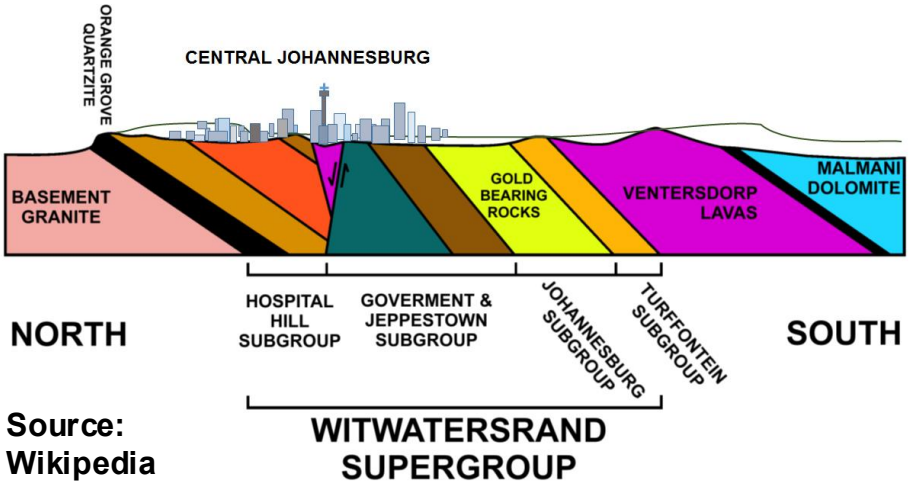


Frimmel 2005



Brandl 2006

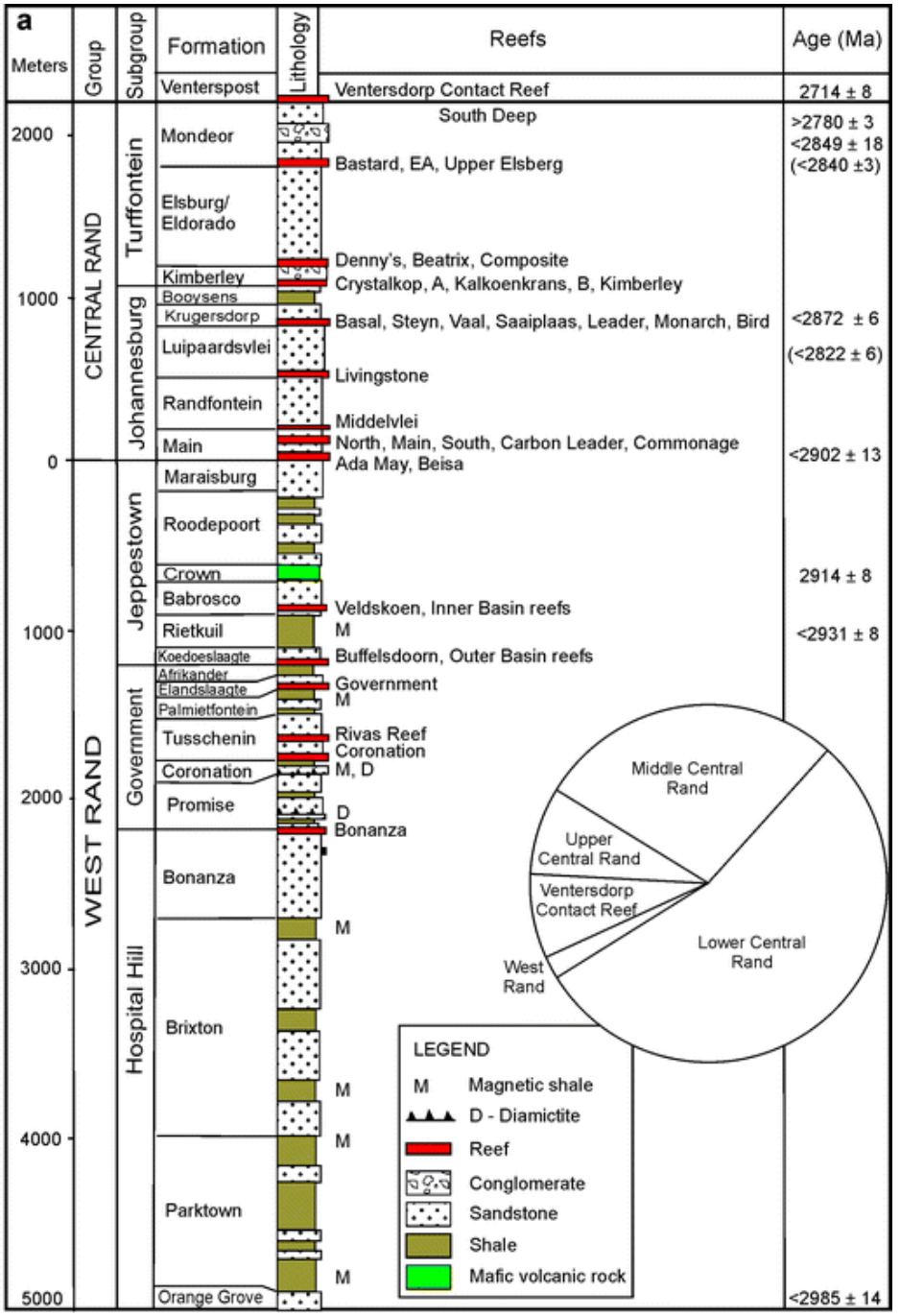
Witwatersrand gold deposits
Stratigraphy



Source: Wikipedia



© theheritageportal.co.za

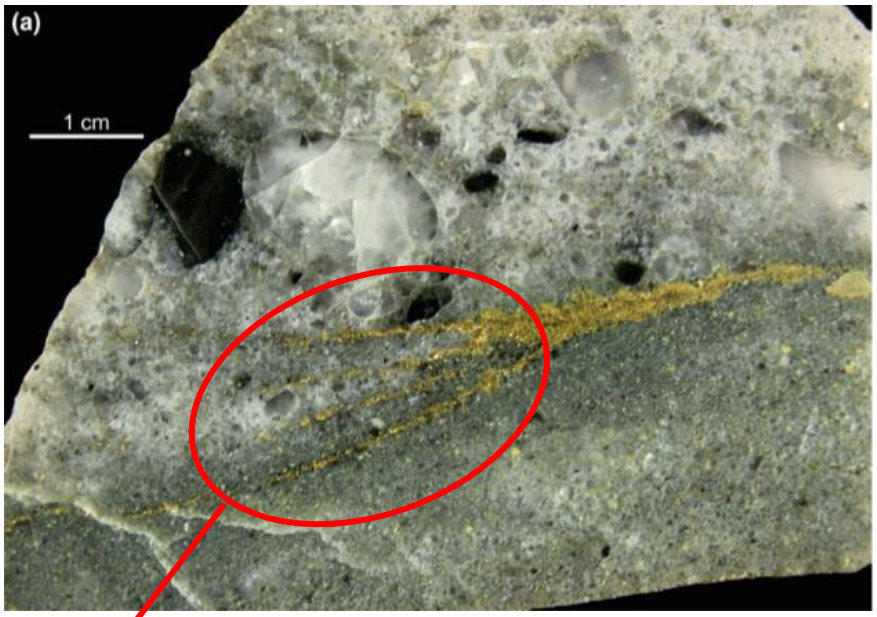


Frimmel 2005

Witwatersrand gold deposits
Mineralogy

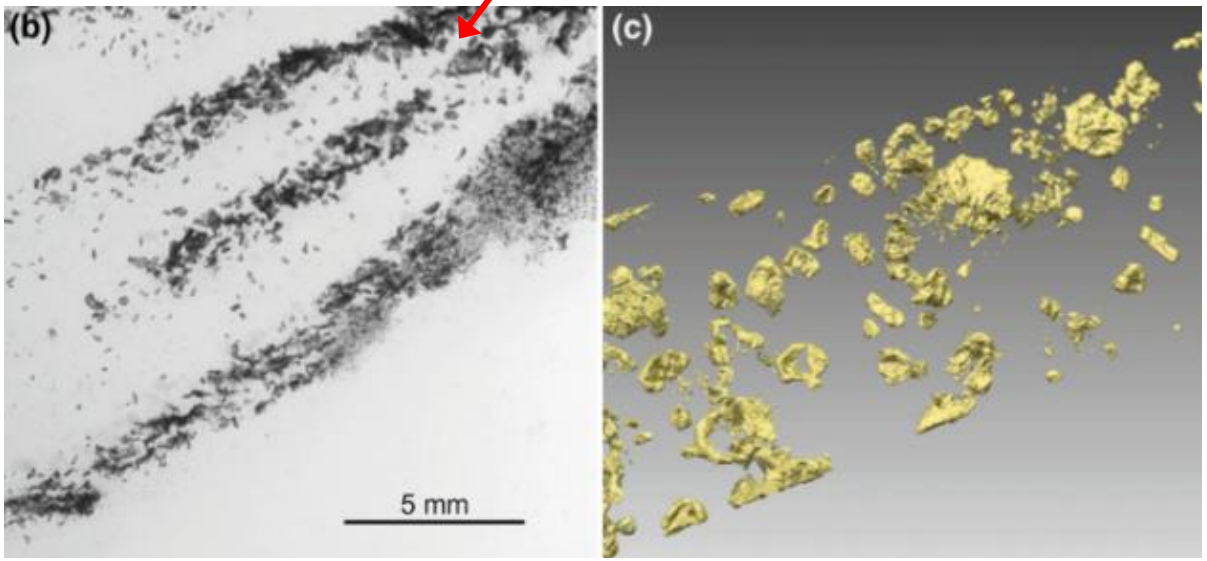


Frimmel 2015



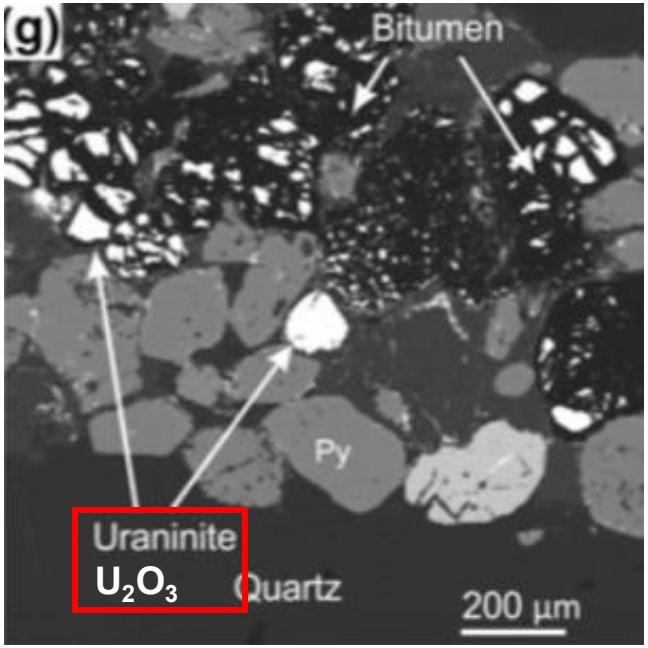
Frimmel 2019

- Conglomerate ‘reefs’ 0.2-4m thick
- Channel fills



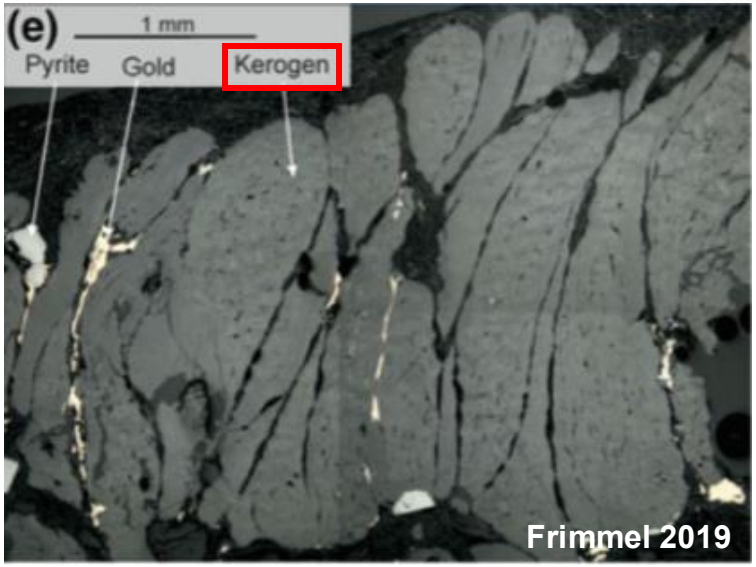
Frimmel 2019

Witwatersrand gold deposits
Mineralogy

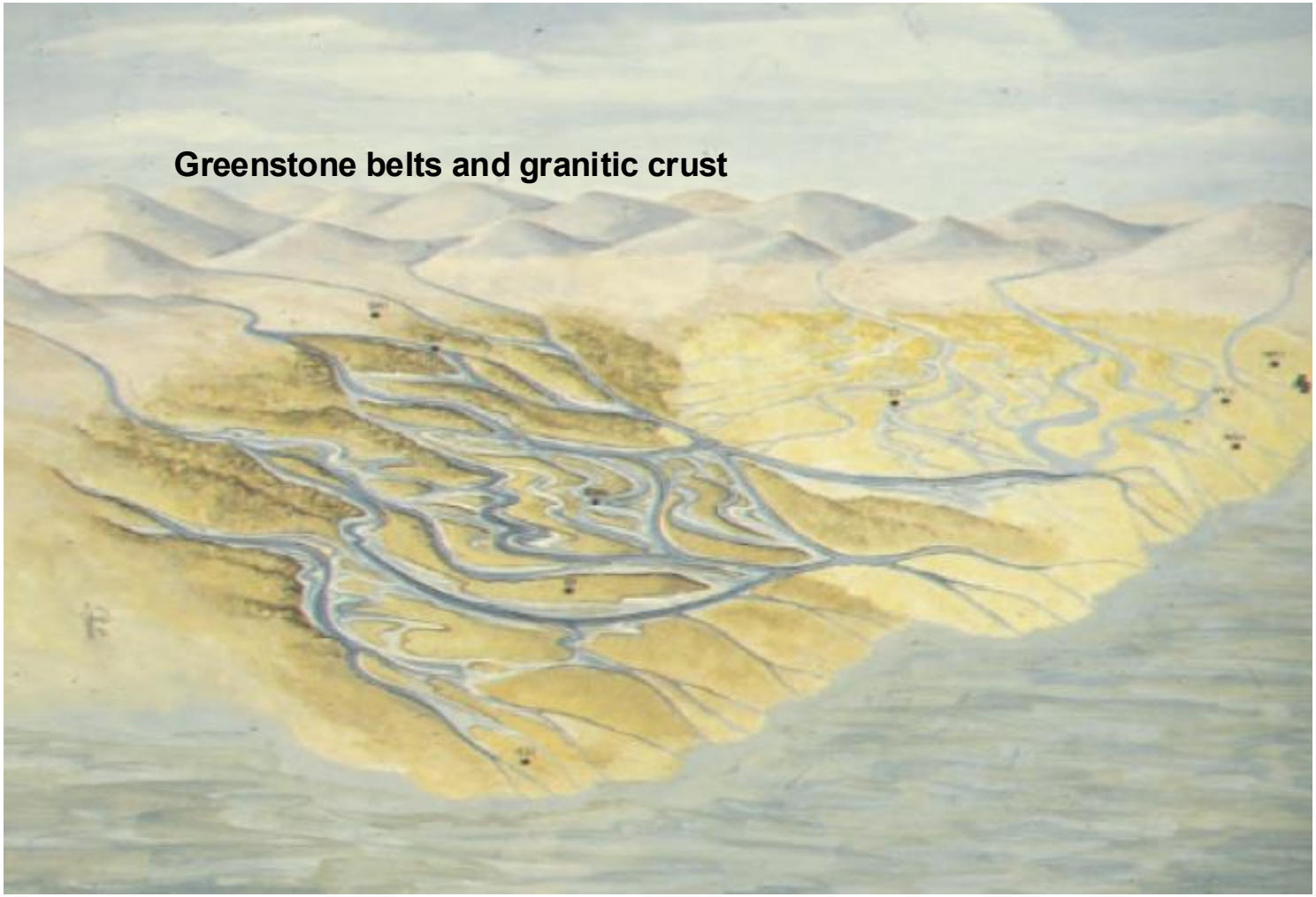


Frimmel 2019





Witwatersrand gold deposits
Depositional system



Frimmel 2005

Witwatersrand gold deposits
Depositional system





© amusingplanet.com

- Braided fluvial system

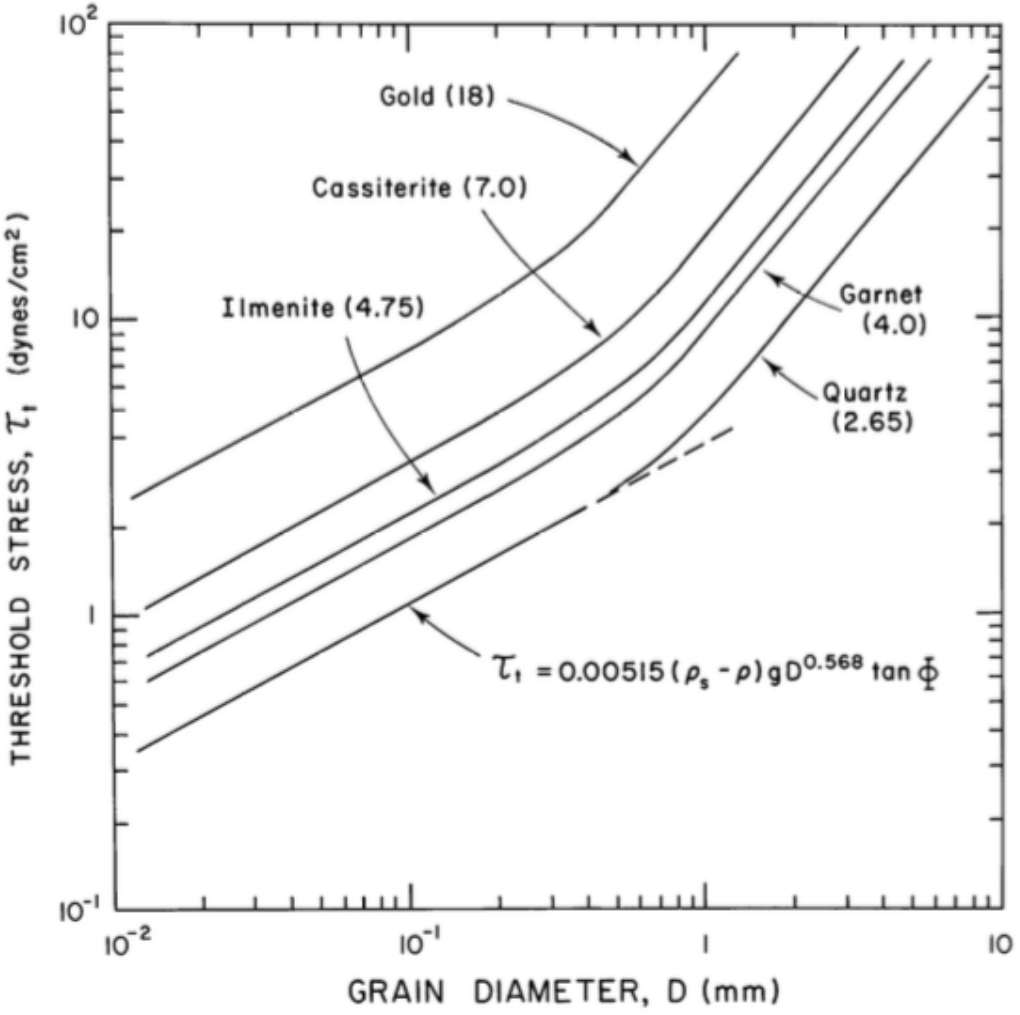
Video braided system

Sediment transport
Application to placer deposits



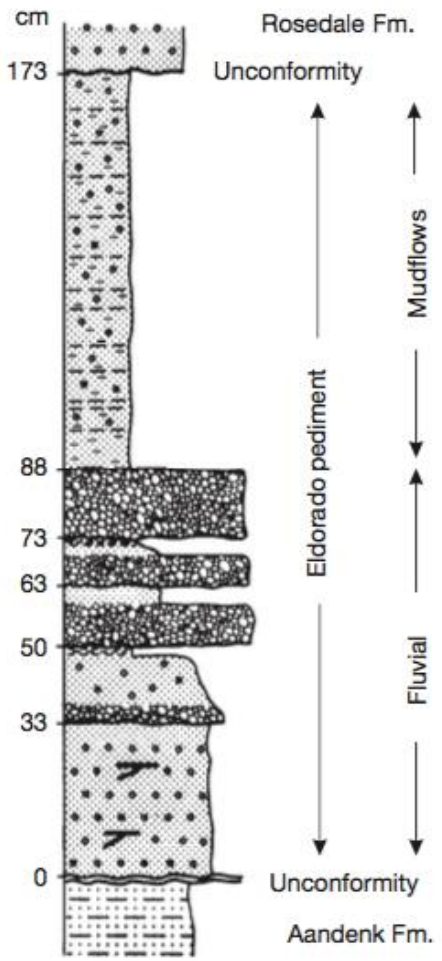
© Prospecting and Mining Journal

$$V = \frac{(\rho_1 - \rho) \times g \times d^2}{18 \times \mu}$$



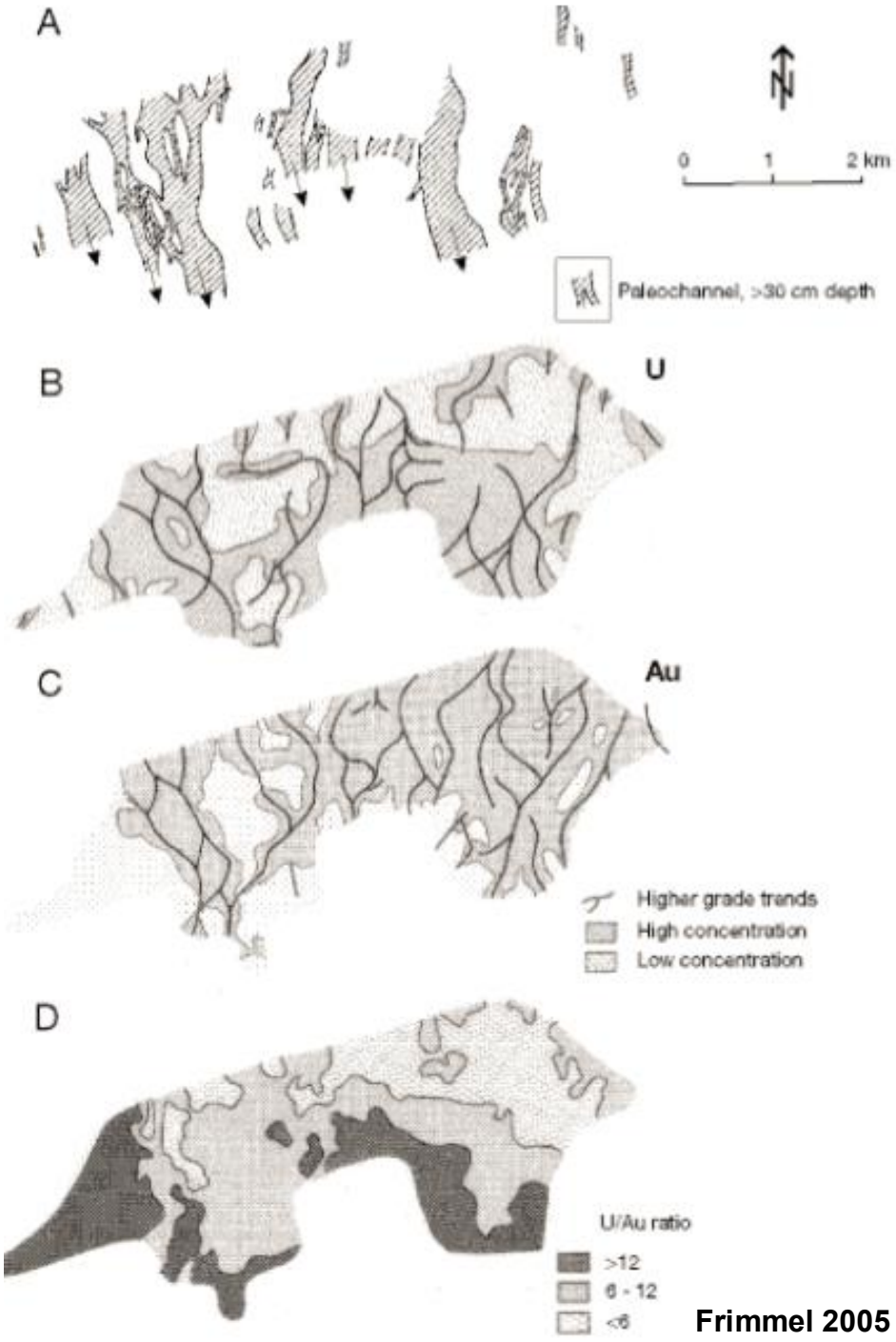
Komar 2007

Witwatersrand gold deposits
Gold accumulation



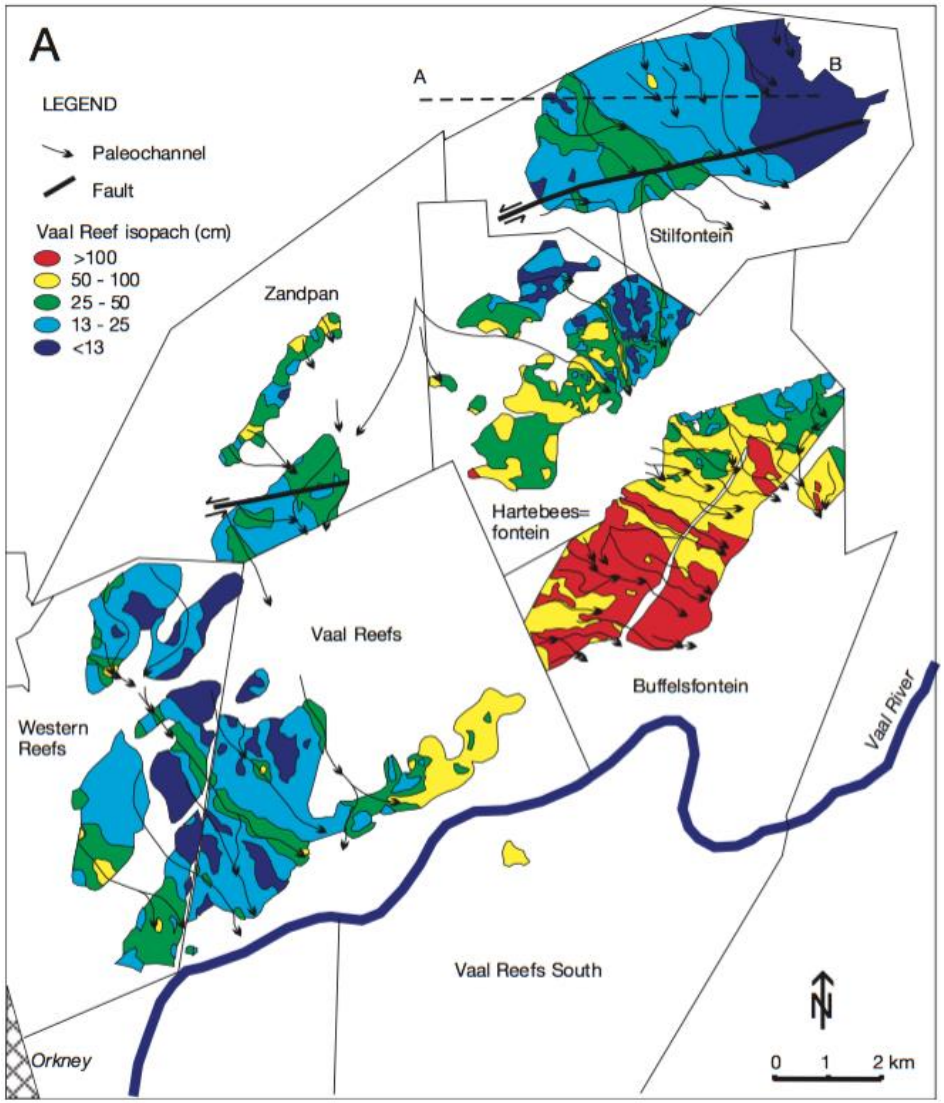
Frimmel 2005

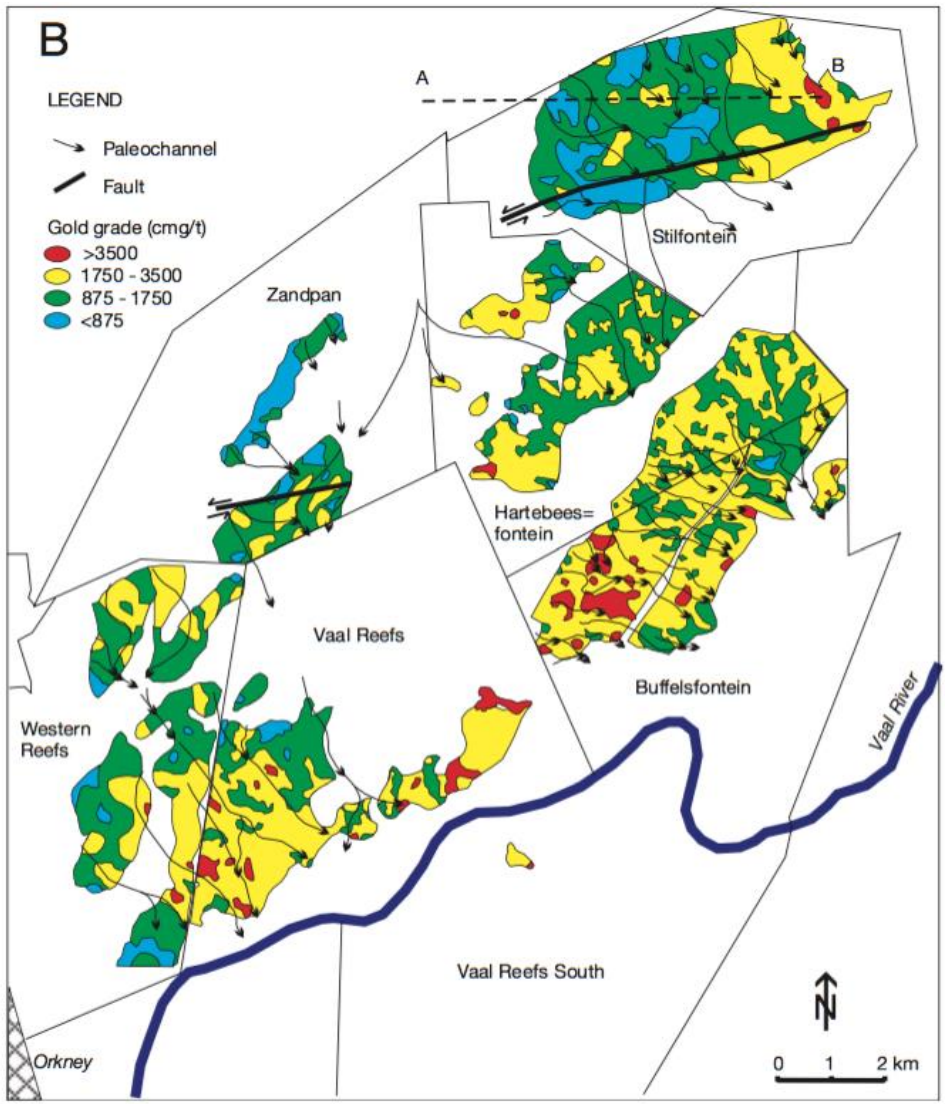
- Grades <25g/t, average ~5g/t (economic 4-10g/t)



Frimmel 2005

Witwatersrand gold deposits
Gold accumulation

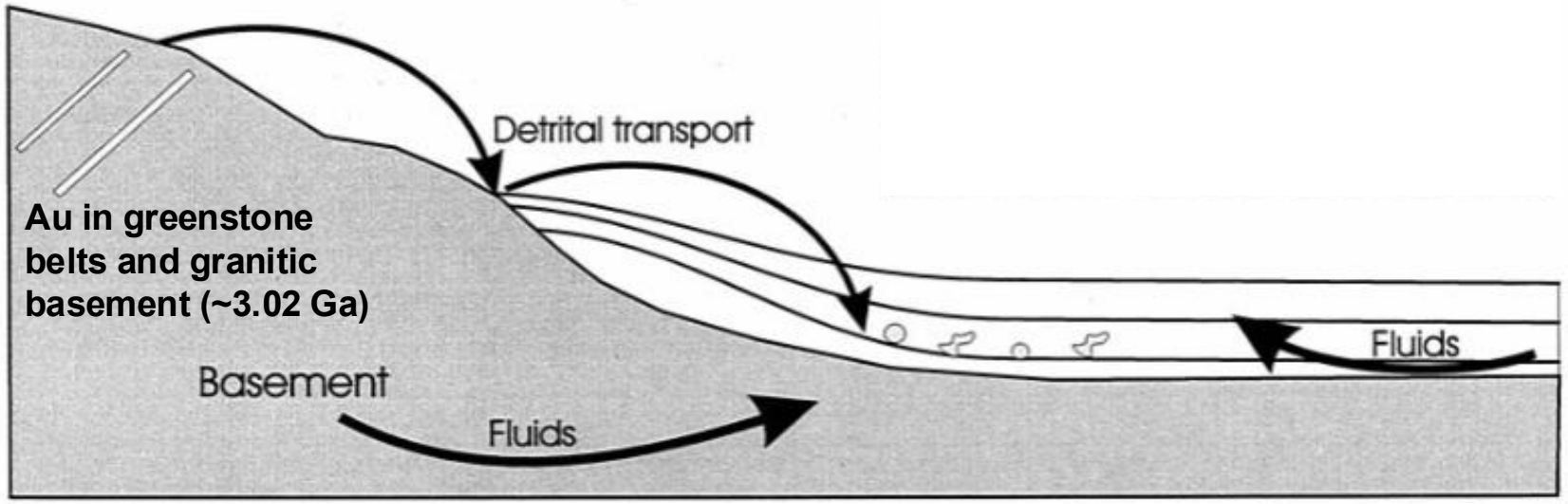




Frimmel 2005

Placer gold deposits
Witwatersrand genesis

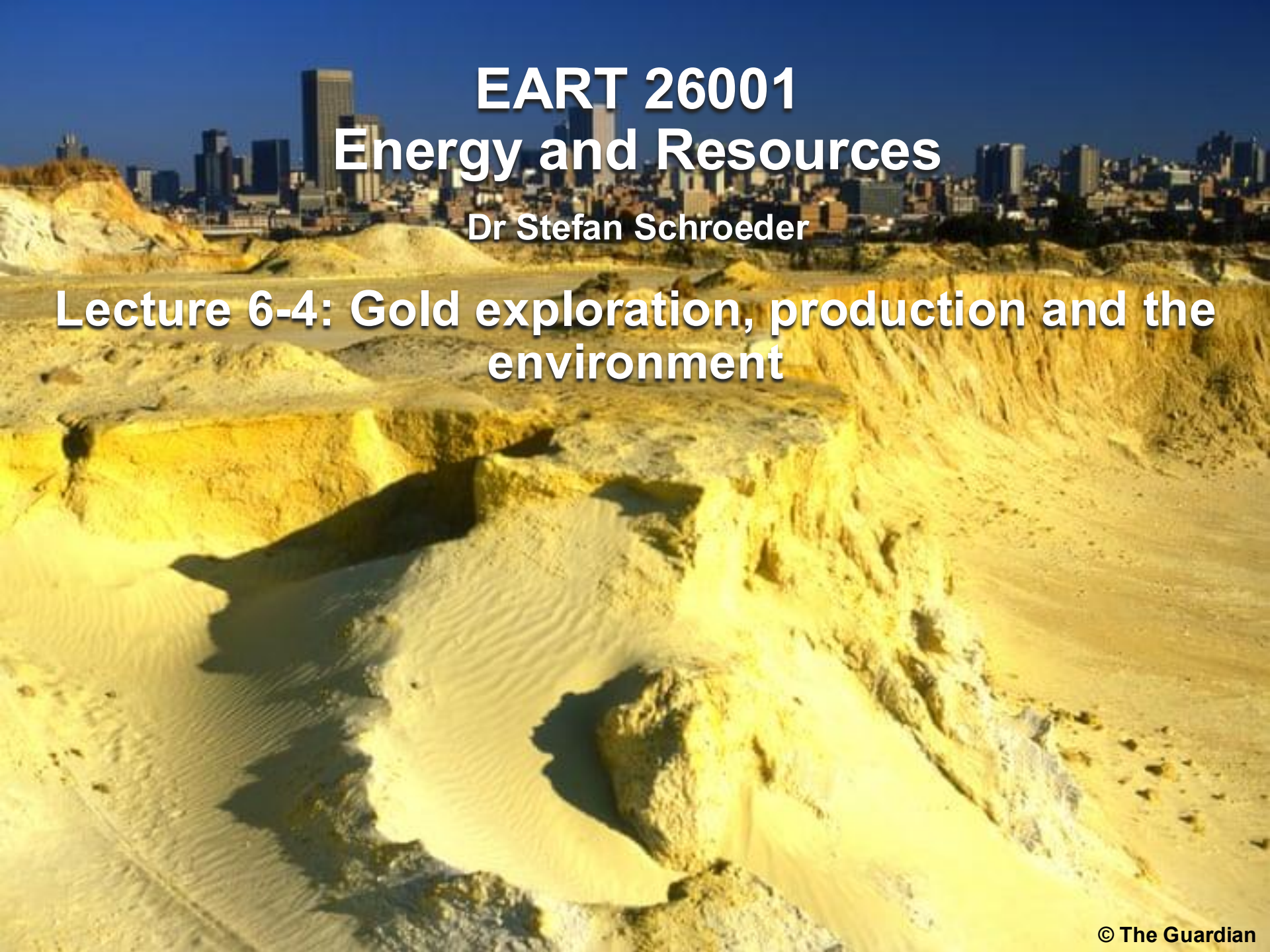
Modified placer model with hydrothermal modification



Kirk 2001

Placer gold deposits
Summary

- Placers generally form where sediment transport allows separation of heavy minerals from other sedimentary grains – gold, ilmenite, magnetite, diamonds etc.
- Understanding sedimentary systems and sediment transport processes are essential in exploration and evaluation of placer deposits
- The Witwatersrand goldfields are one of the largest paleoplacers in the world, formed in a stable basin with delivery of gold-rich sediment from juvenile, gold-rich crust (cf Barberton). Low atmospheric oxygen further allowed for preservation of detrital U and pyrite. These conditions were not replicated later in Earth's history.



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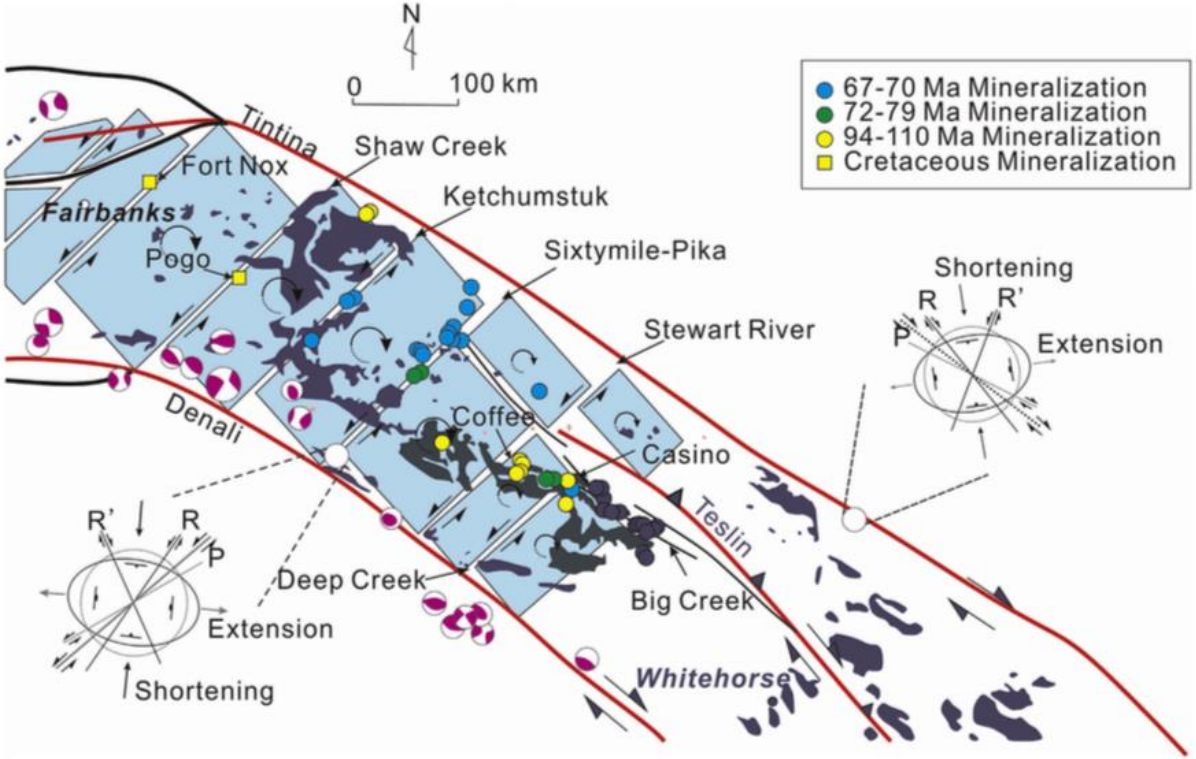
Dr Stefan Schroeder

Lecture 6-4: Gold exploration, production and the environment

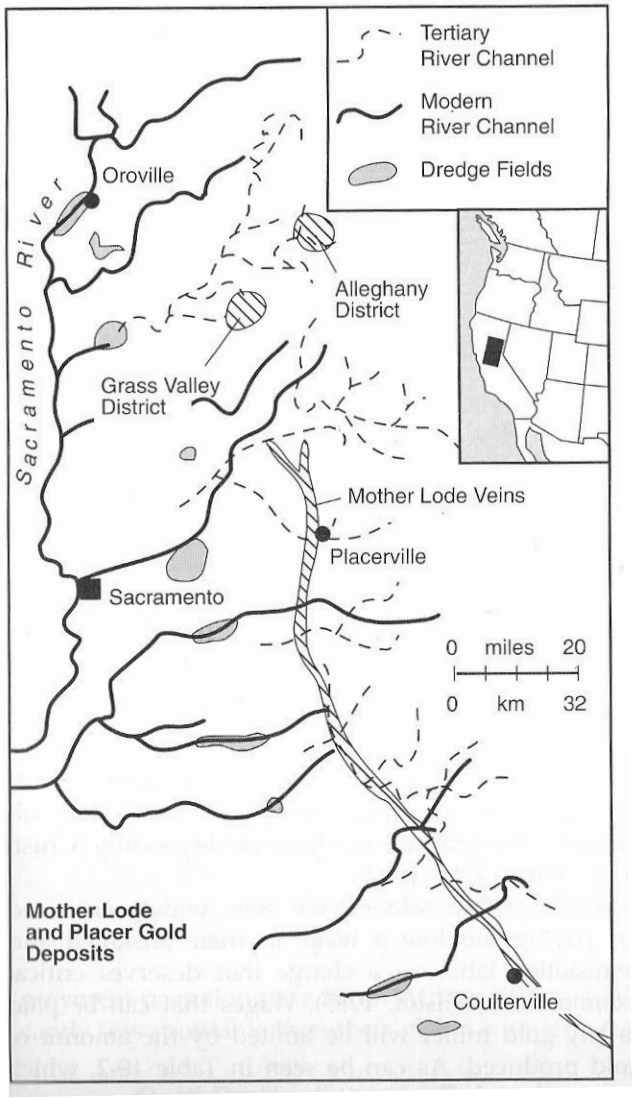
© The Guardian

Gold exploration
Exploration techniques

- Geochemical mapping: trace placer deposits back to source (vein)
- Structural mapping: define relationship between deposits and regional/local structure

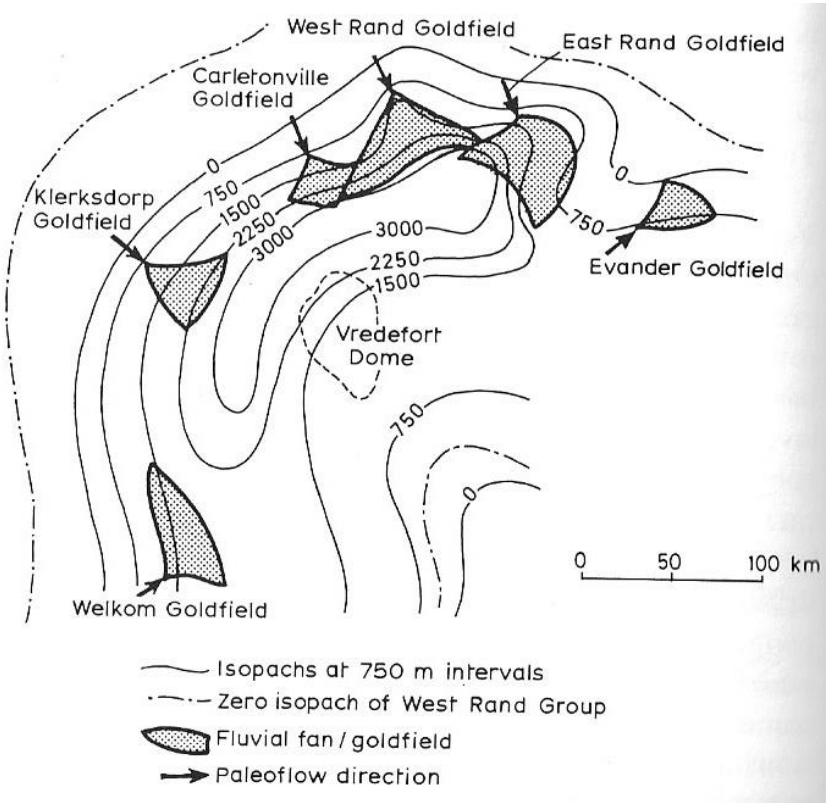


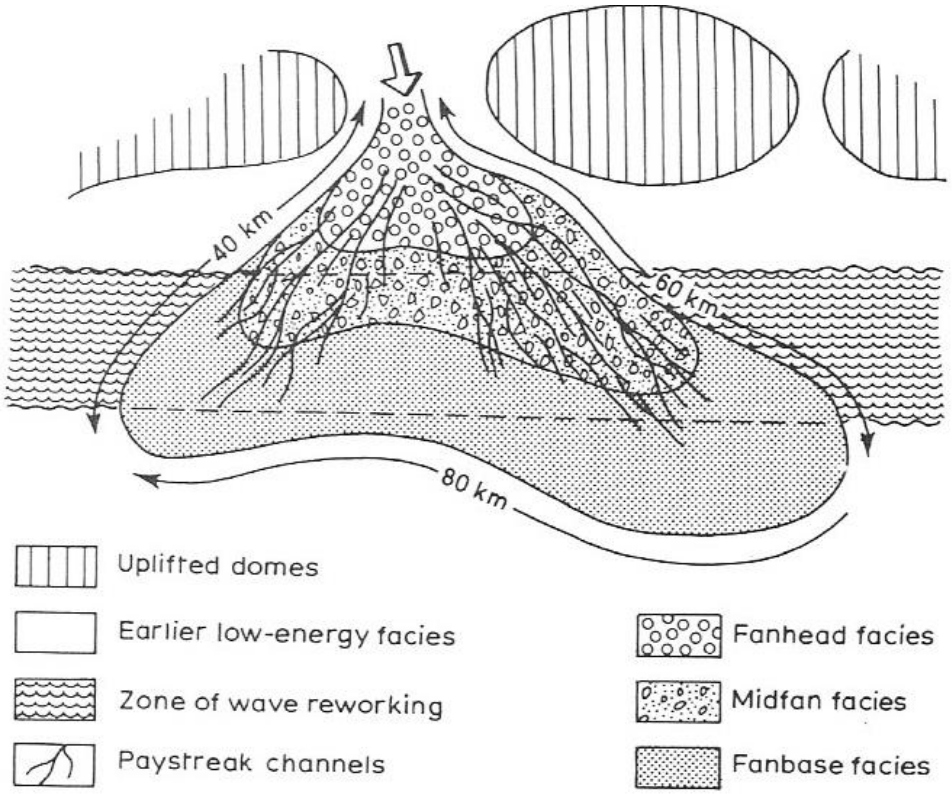
Sanchez 2014



Kesler 1994

Gold exploration
Exploration techniques



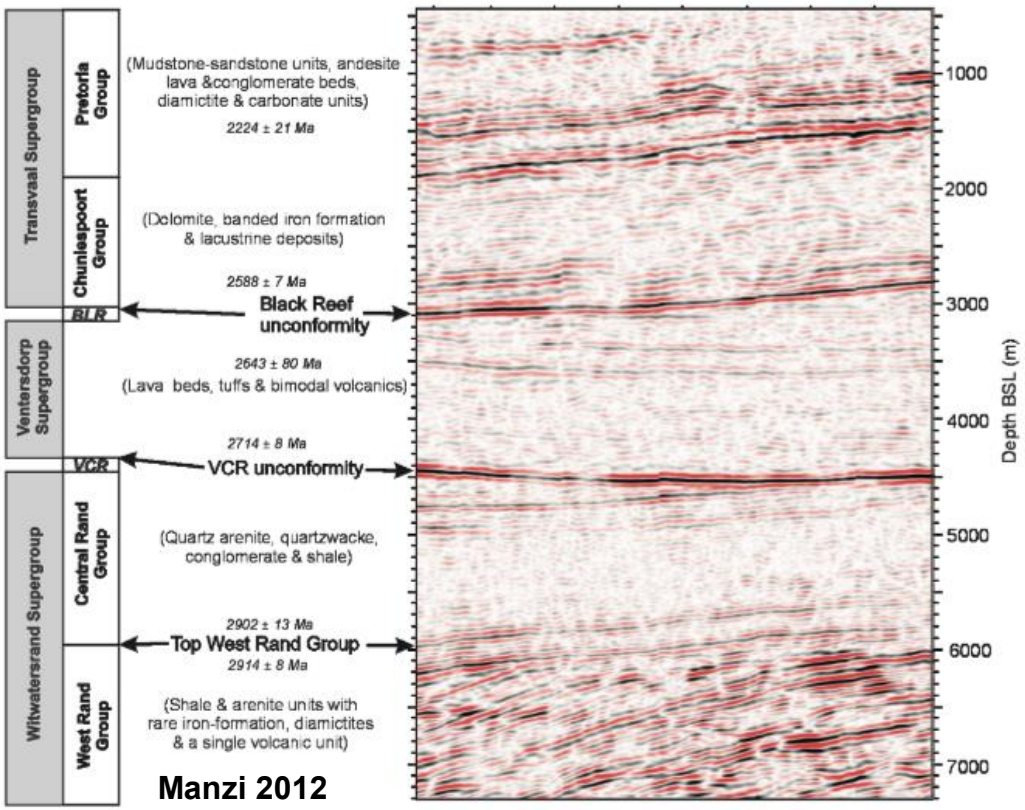


Edwards 1986

- **Depositional system and its controls in placer deposits**

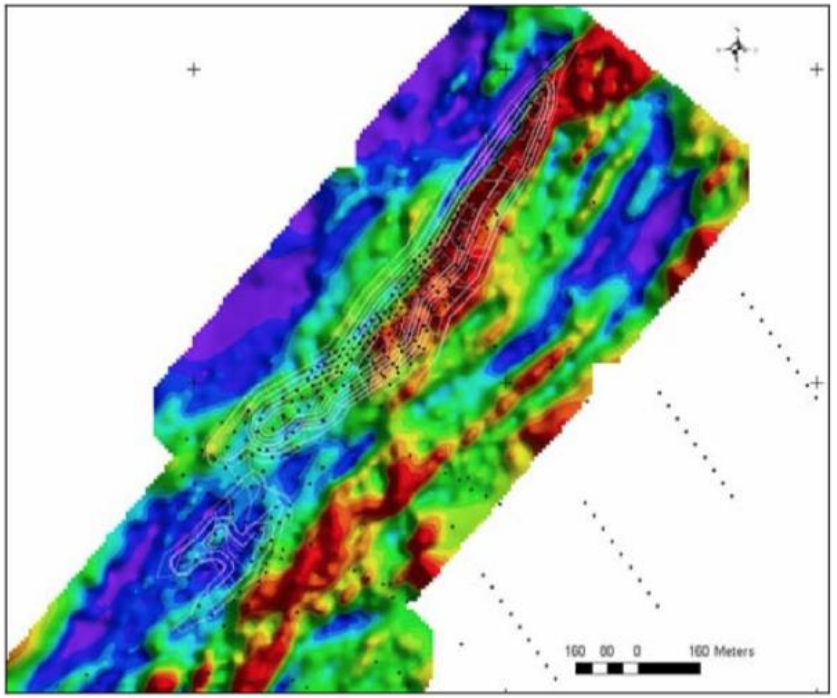
Gold exploration
Exploration techniques

Seismic



Manzi 2012

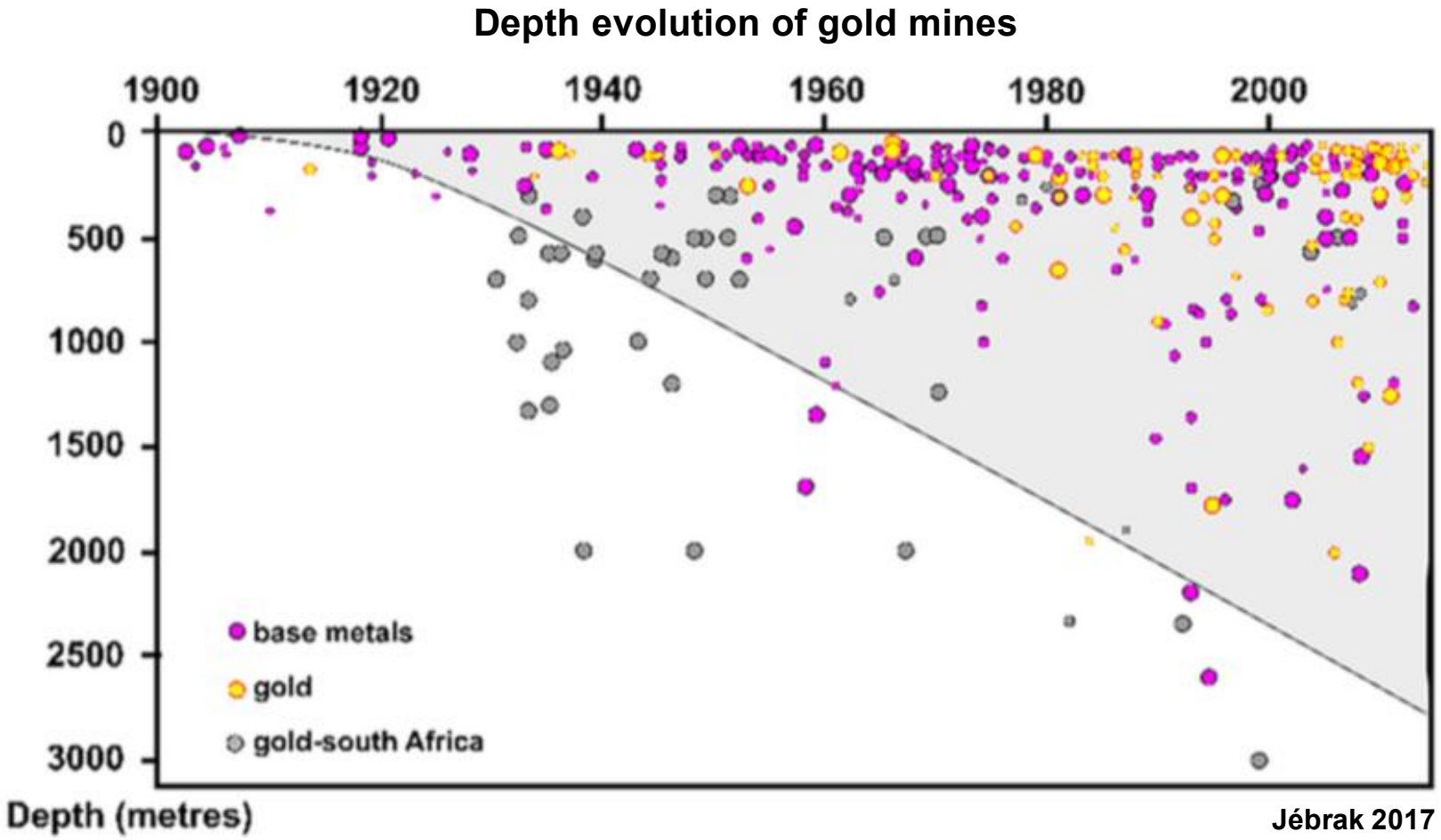
Resistivity



Tsiboah 2009

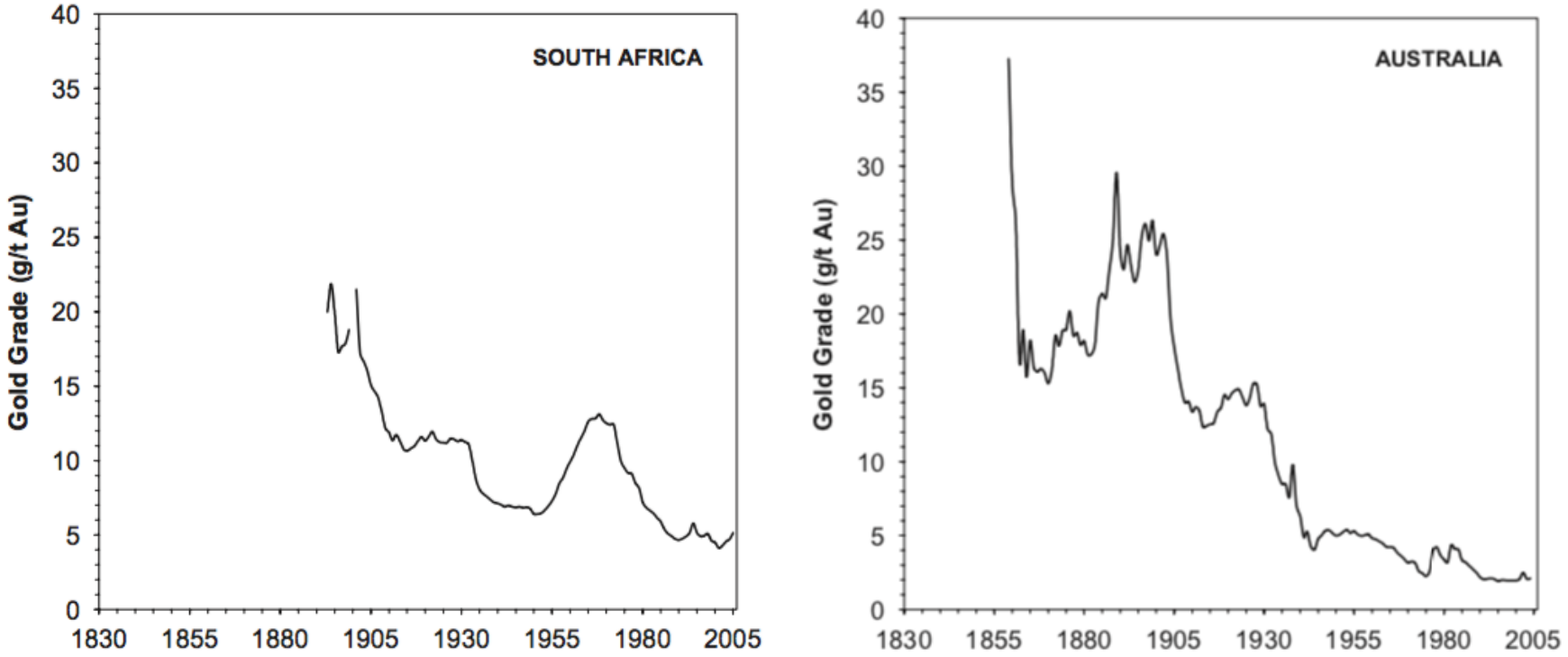
- **Geophysics:**
  - **Magnetics – magnetic minerals (Wits), major structural features**
  - **Resistivity – silica, sulfide content**
  - **Gravity – major structures**
  - **Seismic – major structures**
- **…nothing beats drilling**

Gold production
Mining trends



Gold production
Mining trends

Grade evolution of gold mines



Mudd 2007

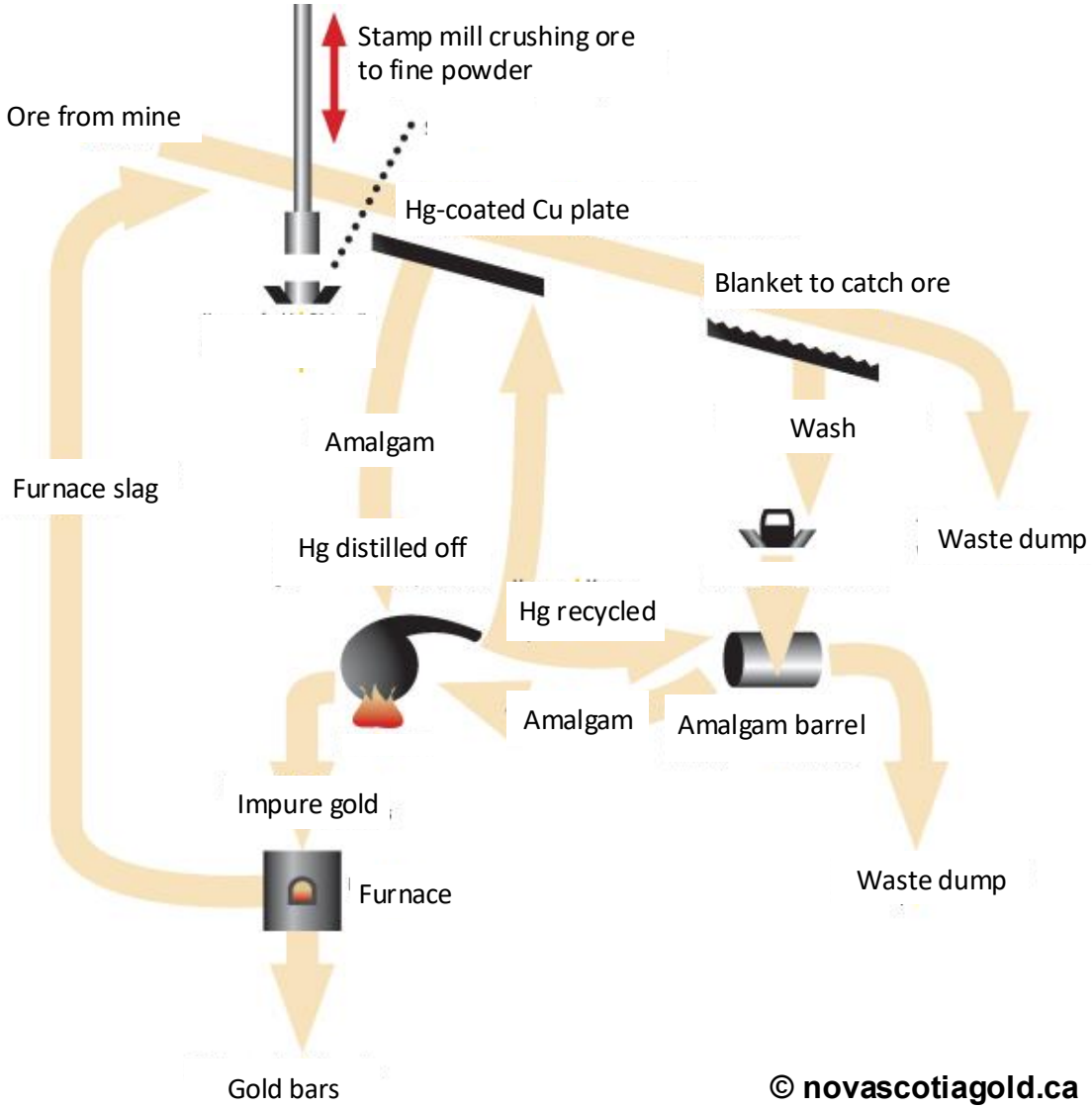
- Current estimates for economic viable gold: 4-10g/t underground, 1-4g/t open pit
- Potential for recycling tailings

Gold production
Beneficiation methods – mercury amalgamation

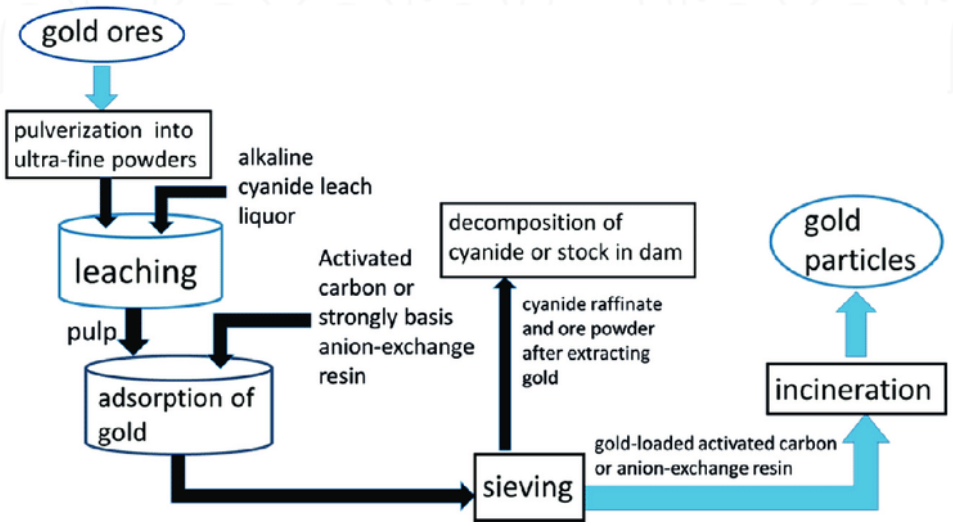
Amalgamation video

- Gold dissolves in Hg to form amalgam
- Amalgam is boiled or dissolved in acid to drive off Hg
- Solid gold powder can be melted into ingots

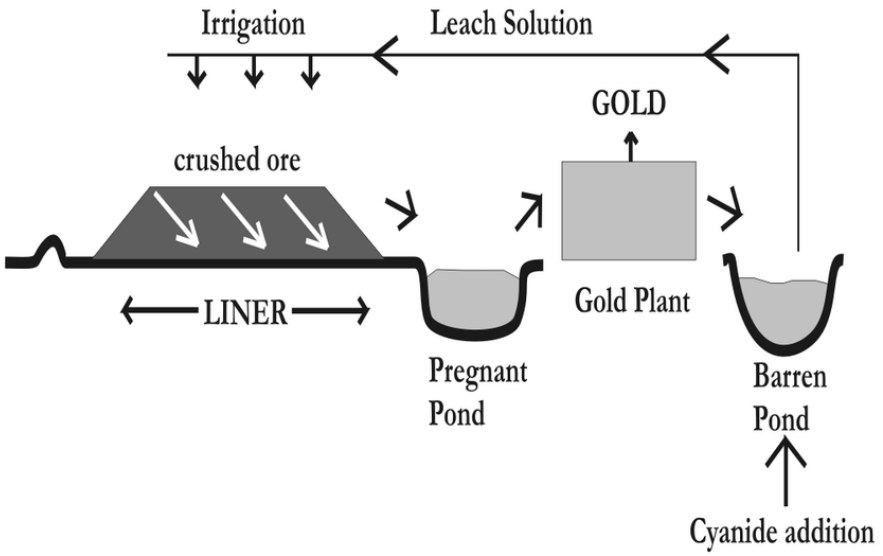
- Process useful for free gold without As, S
- Possibly one of earliest processes, still used in small-scale artisan mining
- Release of poisonous Hg



Gold production
Production methods - cyanidation



Inoue 2019



Mogessie 2002

- Gold dissolved in KCN/NaCN solution:

  $4Au + 8KCN + O_2 + 2H_2O = 4KAu(CN)_2 + 4KOH$

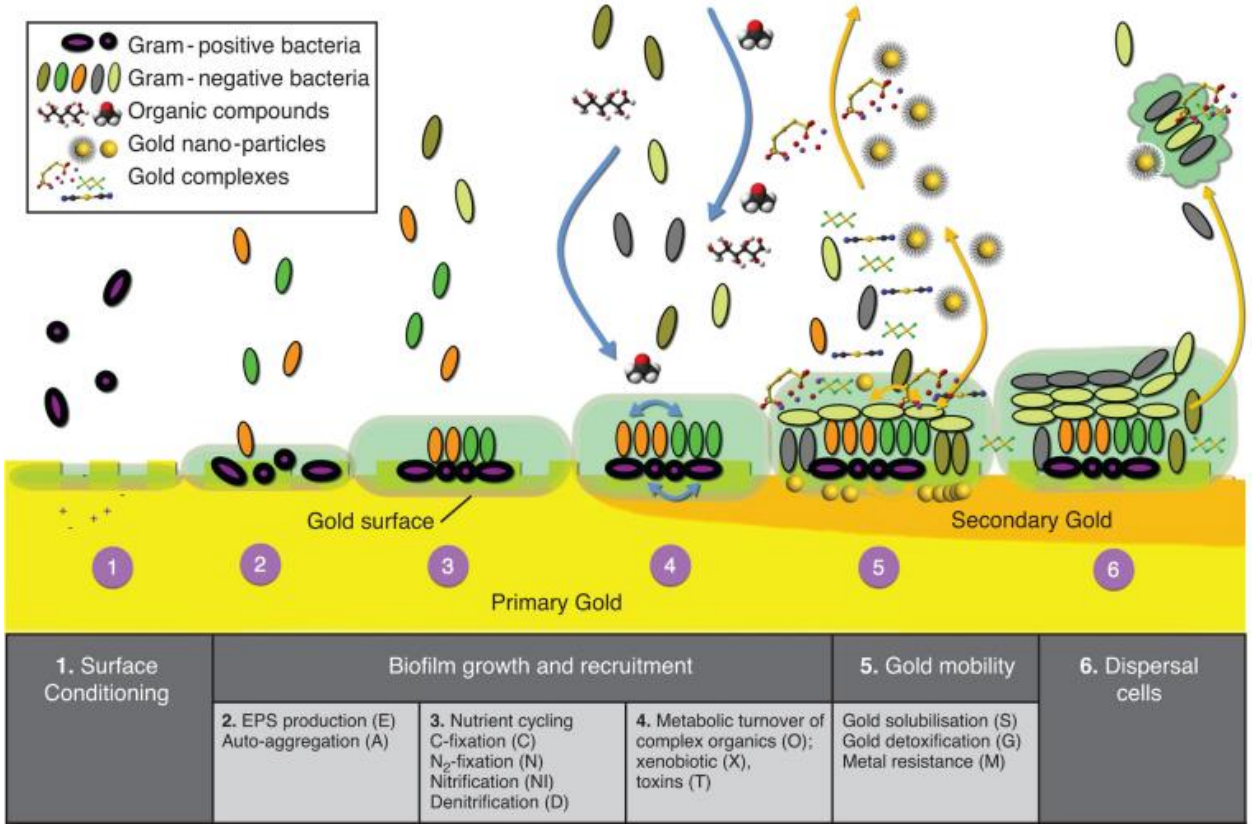
- Recovery by addition of Zn or activated C

- Current industry standard
- Higher recovery than amalgamation, produces metallic gold
- Useful for low concentrations of elements binding with cyanide (As, Zn…) and low S
- Cyanide is poisonous and has polluted groundwaters near gold mines

Gold production
Other production methods

- **Roasting for gold associated with sulfides:**
  - **pyrite is converted to Fe-oxide, Au can be leached with cyanide**
  - **acid mine drainage risk ($4FeS_2 + 15O_2 + 8H_2O \rightarrow 2Fe_2O_3 + 8SO_4^{2-} + 16H^+$), As gas**

- **Bioleaching:**
  - **oxidizes sulfides and allows heap leaching**
  - **can also recover elemental gold directly**



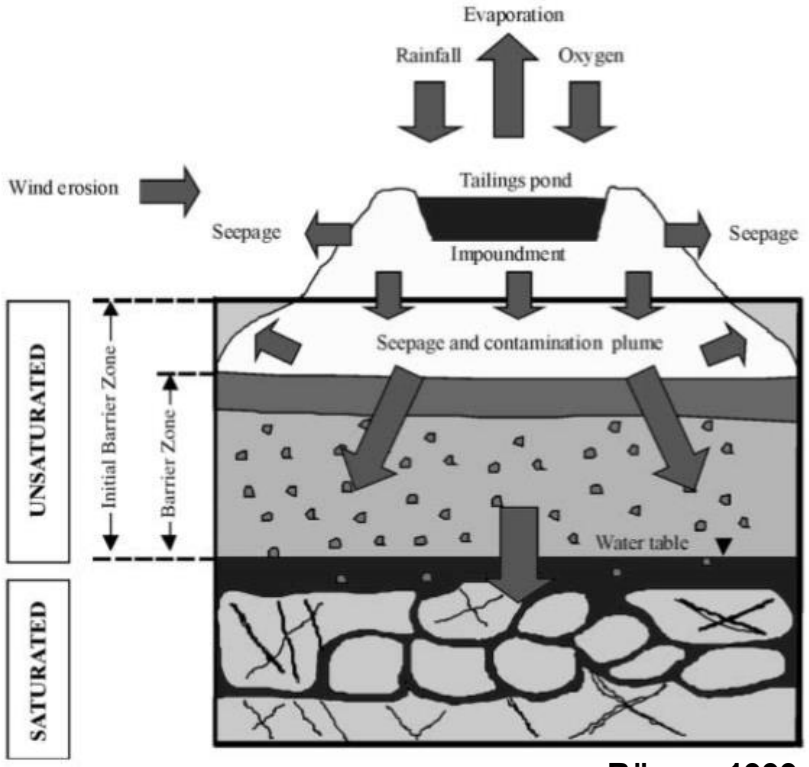
Rea 2016

Gold and its wider impact
Environmental impact

- All beneficiation methods (except bioleaching) have potentially severe negative side effects:
  - Acid mine drainage from roasting
  - Pollution of groundwaters and poisoning of humans/livestock by cyanide and mercury
  - Mine tailings are often left unprotected, letting rain leach remaining chemicals into groundwater, or letting wind carry particles
  - Witwatersrand tailings contain U: low-level radioactive waste
- Recycling of tailings still leaves contaminated footprint
- Greenhouse gas emissions



© D Tang/Ecologies of Gold

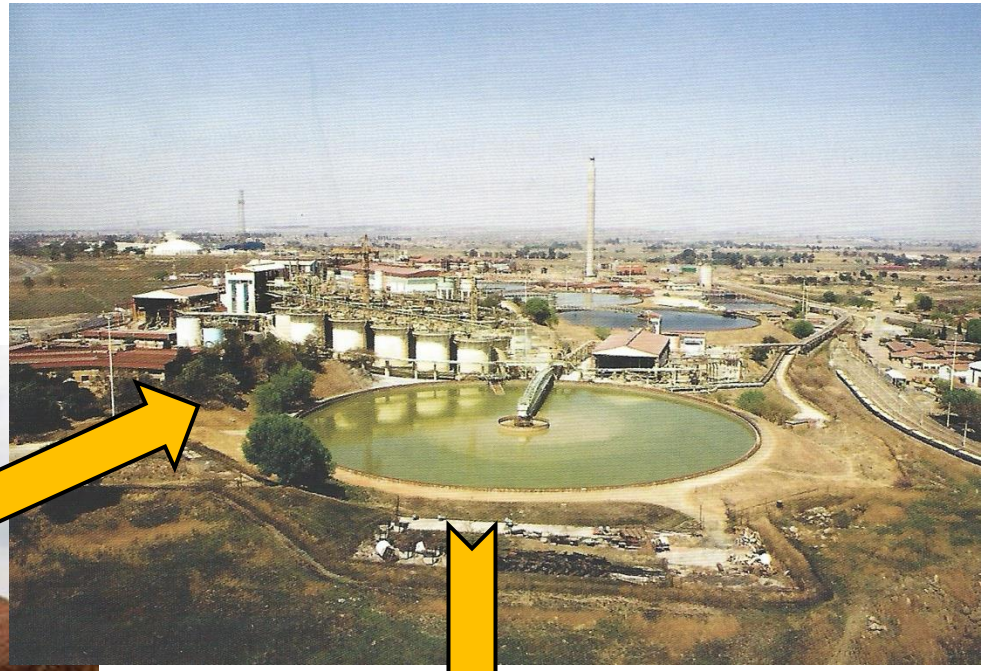


Rösner 1999

Gold and its wider impact
Tailings recycling



© A Watkins/Ecologies of Gold



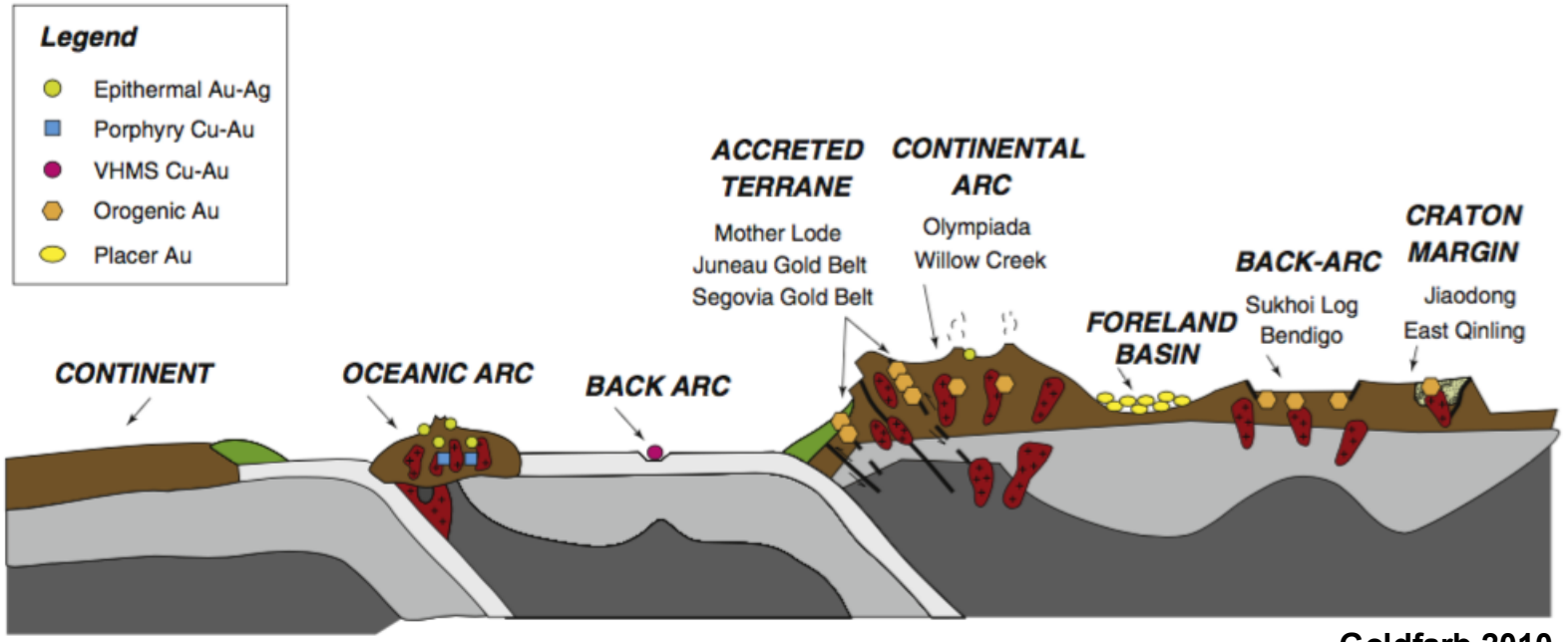
© D Daws/Mining Weekly



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Gold deposits
Key learning points week 6

- Gold occurs in a variety of deposit types:
  - Magmatic-hydrothermal: Epithermal, Porphyries
  - Hydrothermal: Orogenic (typically metamorphogenic)
  - Sedimentary: Placers
- Release of metamorphic hydrothermal fluids, movement along fractures, and formation of gold in veins and disseminations is typical for orogenic deposits.
- Placers (modern and paleoplacer) result from density separation of dense gold particles from other grains during sedimentary transport.
- Processing of gold is environmentally very sensitive.



Goldfarb 2010